

Advanced Algorithms

Practice Workbook

Questions only — a companion to the study website

How to use this workbook

Work topic by topic; within each topic the questions run easy → hard, with 2023+ exams first. Solve each question by hand in the blank space beneath it. This workbook is **questions only** — the **solutions** and the **study guides** live on the study website, where you can reveal each solution and track your progress. Look a question up by its label (e.g. **2025B · P1**). Use the bookmarks/outline to jump between topics.

Solutions & guides: <https://liavburger.github.io/advanced-algorithms-exam-prep/>

Randomized — Probability Basics & Quicksort

13 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · Probability Basics & Quicksort

2023A · P1 (c) **MEDIUM** · 2023 Moed A exam

T/F: $BPP(1/8, 1/4) = BPP(2^{-n}, 1 - 2^{-n})$; error amplification via Chernoff bound.

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(c) $BPP(\frac{1}{8}, \frac{1}{4}) = BPP(2^{-n}, 1 - 2^{-n})$
(n is the input length)

— YOUR WORK —

T/F: $BPP(1/2, 1/2) = BPP(1/3, 2/3)$ (false; the trivial class contains all languages).

Problem 1 (4×6=24 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(c) $BPP(1/2, 1/2) = BPP(1/3, 2/3)$.

— YOUR WORK —

Priority 2 — Homework · Probability Basics & Quicksort

hw2025s1 · P1 (a) **MEDIUM** · HW 2025 · Set 1

Show ZPP equals the class with a poly-time 'I don't know' algorithm, using Markov + geometric-distribution expectation

Part a

We show both directions of the equivalence. Let $A \in ZPP$. Let M be a zero-error randomized algorithm that decides A whose running time is polynomial in expectation. Let T be a random variable that denotes the running time of M , so that on every input of length n , $\mathbb{E}[T] \leq f(n)$ for some polynomial $f(n)$.

We'll show that A has an algorithm M' with properties (i)–(iv) as stated in the question. Let M' be an algorithm that, on input x of length n , simulates M for at most $10 \cdot f(n)$

— YOUR WORK —

Show RP subset of BPP and $coRP$ subset of BPP by probability amplification

Part b

Recall that $BPP = BPP(1/3, 2/3)$. Let $A \in RP$. Then there's an algorithm M such that if $x \notin A$, M accepts x with probability 0, and if $x \in A$, M accepts x with probability at least $1/2$. Consider an algorithm M' which runs A twice, each time with fresh and independent

— YOUR WORK —

Prove $ZPP = RP \text{ intersect } coRP$

Part c

We may use the definition of ZPP from part (a). Suppose $A \in ZPP$. Let M be an algorithm for A with properties (i)–(iv) from part (a). Consider an algorithm M' which is identical to A except that if M says “I don’t know”, M' rejects. M' runs in polynomial time. If

— YOUR WORK —

Priority 3 — Pre-2023 Exams · Probability Basics & Quicksort

2017A · P1 (a) **EASY** · 2017 Moed A exam

Solve recurrence $T(n)=3T(n/4)+\Theta(n)$ for a divide-and-conquer running time (master theorem).

- a. An algorithm divides an input of size n into 3 subproblems, each of size $n/4$. This takes $\Theta(n)$ time. The subproblems are solved recursively, and combined into a solution of the original problem in $O(n^{2/3})$ time. Problems of size $n < 500$ are solved directly in n^3 time. What is the asymptotic running time of the algorithm?

— YOUR WORK —

Solve two-variable recurrence $T(m/2, n/2) + c(m+n)$ via geometric sum, giving $\Theta(m+n)$.

- b. A divide-and-conquer algorithm takes as input an m -by- n matrix M with $m \leq n$. The divide step requires $\Theta(n + m)$ time, and produces a $\lfloor m/2 \rfloor$ -by- $\lfloor n/2 \rfloor$ matrix M' , which is handled recursively. Any 1-by- n matrix is handled directly in $\theta(n)$ time. The combine step also requires $\Theta(m + n)$ time. What is the asymptotic running time of the algorithm?

— YOUR WORK —

True/False: $E[2x-y]=2E[x]-E[y]$ for dependent random variables (linearity of expectation).

- c. True or False: "Let Ω be some sample space. Let x, y be two random variables (not necessarily independent). It is always true that $E[2x - y] = 2E[x] - E[y]$ ". Give a short explanation.

— YOUR WORK —

Solve recurrence $T(5n/6) + T(n/10) + \Theta(n^2)$ (top level dominates), giving $\Theta(n^2)$.

- b) An algorithm breaks a problem of size n into 2 subproblems, one of size $5n/6$ and the other of size $n/10$. This process takes $\Theta(n^2)$ time. It then solves each subproblem recursively, and combines the solutions in $\Theta(n)$ time. What is the asymptotic running time of the algorithm?

— YOUR WORK —

Does the expected-runtime analysis of randomized selection apply to the worst-case input? (Yes.)

4 . (25 points) Recall the randomized selection algorithm we saw in class. The algorithm chooses a pivot element uniformly at random, partitions the array into elements smaller than the pivot and larger than the pivot in $\theta(n)$ time, and recurses on the appropriate part. In class we analyzed the expected running time of the algorithm.

a) (2 points) Does the analysis of the expected running time apply to the worst case input array? (Yes / No)

— YOUR WORK —

What is the expectation in the randomized selection analysis over? (The random pivot choices.)

b) (3 points) What is the expectation over?

— YOUR WORK —

Prove the recurrence $E[T(n)] \leq (2/n) \sum E[T(k)] + \Theta(n)$ for randomized selection.

- c) (5 points) In class, we denoted by $T(n)$ the r.v. for the running time of the selection algorithm on an array of length n . Prove that $E[T(n)]$ satisfies the following recurrence:

$$E[T(n)] \leq \frac{2}{n} \sum_{k=\frac{n}{2}}^{n-1} E[T(k)] + \theta(n)$$

You may assume that the time for partition is $\theta(n)$, and may ignore rounding issues.

— YOUR WORK —

Prove by induction that $E[T(n)] = O(n)$ for randomized selection using the given recurrence.

- d) (7 points) Use the recurrence stated in part (c) to prove by induction that $E[T(n)] = O(n)$. You may use the fact that the sum of an arithmetic series is given by the sum of the first and last elements in the series, multiplied by half the number of elements in the series. Make sure you clearly write the inductive assumption, the base case, and the inductive step.

— YOUR WORK —

Randomized — Karger's Min-Cut

3 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · Karger's Min-Cut

2024B · P3 **MEDIUM** · 2024 Moed B exam

Prove a graph has at most $n(n-1)/2$ minimum cuts using Karger's success-probability analysis.

Problem 3 (12 points)

Let G be an undirected graph.

Prove that the number of minimum cuts in G is at most $\frac{n(n-1)}{2}$.

(Hint: recall the analysis of the success probability of Karger's algorithm)

— YOUR WORK —

Priority 2 — Homework · Karger's Min-Cut

hw2026s1 · P3 **MEDIUM** · HW 2026 · Set 1

Use Karger's min-cut success bound to prove a graph has at most $n(n-1)/2$ minimum cuts

Problem 3

Let (A_0, B_0) be a fixed minimum cut. As we've seen in class, the probability that Karger's algorithm returns (A_0, B_0) is at least $\frac{2}{n(n-1)}$. This is true for every minimum cut in G , and the probability of the algorithm returning some minimum cut is obviously at most 1.

— YOUR WORK —

Priority 3 — Pre-2023 Exams · Karger's Min-Cut

2018A · P1 (e) **EASY** · 2018 Moed A exam

Is the randomized min-cut algorithm Monte Carlo or Las Vegas? (Monte Carlo.)

e) Is the randomized algorithm for minimum cut we saw in class Monte Carlo or Las Vegas?

— YOUR WORK —

Randomized — Concentration Bounds

14 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · Concentration Bounds

2025B · P1 (c) **EASY** · 2025 Moed B exam

T/F: at most half the students score at least twice the average grade (Markov's inequality).

Problem 1 (4×8=32 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(c) At most half of the students in this exam will receive a grade that is at least twice the average grade.

— YOUR WORK —

Randomized coloring achieving discrepancy $\leq C\sqrt{n \log m}$ whp, via Chernoff and union bound over m sets.

Problem 6 (18 points)

In the **Small Discrepancy** problem, the input is a family $\mathcal{F}$ of subsets of $\{1, \dots, n\}$. For a coloring h of the elements $\{1, \dots, n\}$ in red and blue, the **discrepancy** of a set $S \subseteq \{1, \dots, n\}$ under h , denoted $\text{Disc}_h(S)$, is the different in absolute value between the number of red elements in S and the number of blue elements in S .

The discrepancy of $\mathcal{F}$ under the coloring h is $\max_{S \in \mathcal{F}} \text{Disc}_h(S)$.

Give an efficient **randomized** algorithm that, given a family $\mathcal{F}$ of m sets, finds (with high probability) a coloring h such that $\text{Disc}_h(\mathcal{F}) \leq C\sqrt{n \log m}$, where C is some absolute constant (you can pick C as large as you want, but it shouldn't depend on n and m).

Randomized algorithm estimating a CNF formula's satisfying-assignment density within $1/10$ w.p. $1-2^{-n}$ via sampling + Chernoff.

Recall that a CNF formula is a formula of the form

$$C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

where every clause C_i is of the form $\ell_1 \vee \ell_2 \cdots \vee \ell_t$, and every literal ℓ_i is a variable or its negation.

For an formula φ with n variables, we define its **density**, denoted $D(\varphi)$, to be the fraction of assignments satisfying φ among the total 2^n assignments, namely,

$$D(\varphi) = \frac{1}{2^n} \cdot |\{x \in \{0, 1\}^n : \varphi(x) = 1\}|$$

Prove that there's a randomized polynomial time algorithm with the following property: given a CNF formula φ with n variables, with probability at least $1-2^{-n}$, the algorithm outputs a number $A(\varphi)$ such that

$$|A(\varphi) - D(\varphi)| \leq \frac{1}{10}.$$

ז מהצורה

צורה $\ell_1 \vee \ell_2 \cdots \vee \ell_t$ או שלילתו.

שתנים, נגדיר את $D(\varphi)$, להיות אחוז בין כל 2^n ההשמות

י שרץ בזמן פולינומי נוסחת CNF φ עם $1-2^{-n}$ הפלט של שמתקיים

Randomized algorithm finding a binary vector at Hamming distance $\geq n/3$ from all of S , via Chernoff + union bound.

Problem 3 (18 points)

Let $u, v \in \{0, 1\}^n$. The **Hamming distance** between u and v , denoted $d(u, v)$, is defined to be the number of coordinates on which they differ.

Give a randomized algorithm that given a set of vectors $S \subseteq \{0, 1\}^n$ of size at most n^{2024} , finds, with probability at least 0.9, a vector $u \in \{0, 1\}^n$ such that for every $v \in S$ it holds that $d(u, v) \geq n/3$.

— YOUR WORK —

Randomized estimator of the fraction of legal 3-colorings within $1/n$ whp, via sampling and Chernoff.

Problem 5 (16 points)

Let $G = (V, E)$ be an undirected graph with n vertices.

A **3-coloring** of the vertices of G is a function $c : V \rightarrow \{1, 2, 3\}$.

c is called **legal** if for every $(u, v) \in E$, $c(u) \neq c(v)$.

The **versatility** of G , denoted $R(G)$, is the percentage of legal 3-colorings among all 3-colorings, that is,

$$R(G) = \frac{|\{c : c \text{ is a legal coloring}\}|}{3^n}.$$

Give a randomized algorithm that runs in polynomial time that given a graph G , with probability at least $1 - 2^{-n}$, outputs a number $A(G)$ such that $|A(G) - R(G)| \leq \frac{1}{n}$.

— YOUR WORK —

Construct a $(C \log n, 1/100)$ -balanced bipartite graph whp: random edges, Chernoff + union bound over all subsets.

Problem 6 (18 points)

Let $G = (L \cup R, E)$ a bipartite graph with n vertices on each side.

For subsets $S \subseteq L, T \subseteq R$ let $E(S, T)$ denote the set of edges between S and T .

G is called (k, ε) -**balanced** if for every $m \geq k$ and for every two subsets $S \subseteq L, T \subseteq R$ such that $|S| = |T| = m$, it holds that $\left| E(S, T) - \frac{1}{2}m^2 \right| \leq \varepsilon m^2$.

Prove that there exists a constant C such that there exists a randomized algorithm that on input n , runs in time polynomial in n , and with probability at least 0.9, its output is a $(C \log n, \frac{1}{100})$ -balanced graph.

— YOUR WORK —

Randomized poly-time algorithm outputting an epsilon-unbiased set of size $O(n/\epsilon^2)$ w.p. 0.99 via random sampling + Chernoff + union bound.

For a subset $\emptyset \neq I \subseteq \{1, 2, \dots, n\}$ and a vector $v \in \{-1, 1\}^n$, let $e_I(v) = \prod_{i \in I} v_i$.
A subset $S \subseteq \{-1, 1\}^n$ is called **ϵ -unbiased** if **for every** $\emptyset \neq I \subseteq \{1, 2, \dots, n\}$,

$$\left| \sum_{v \in S} e_I(v) \right| \leq \epsilon |S|$$

Show that there exists a randomized algorithm that for every n , runs in time polynomial in n , and with probability at least 0.99, outputs an ϵ -unbiased set $S \subseteq \{-1, 1\}^n$ of size $O(n/\epsilon^2)$.

— YOUR WORK —

Priority 2 — Homework · Concentration Bounds

hw2025s1 · P5 **MEDIUM** · HW 2025 · Set 1

Random ± 1 coloring keeps every set's discrepancy $O(\sqrt{n \log m})$ via Chernoff + union bound

Problem 5

The algorithm is very simple: just assign a uniformly random color independently for each element $i \in \{1, \dots, n\}$. We would like to show that with high probability, this random coloring h satisfies the requirements. For every i , let X_i be a random variable which is 1 if i is colored red, and -1 if i is colored blue. Note that by our algorithm, $\mathbb{E}[X_i] = 0$, and the

— YOUR WORK —

Sample s random ± 1 vectors so every character e_I has small bias, via Chernoff + union bound

Problem 6

The algorithm picks $v_1, \dots, v_s$ independently and uniformly at random from $\{-1, 1\}^n$. Suppose $v \in \{-1, 1\}^n$ is picked at random. For every $\emptyset \neq I \subseteq \{1, 2, \dots, n\}$, the random variable $e_I(V)$ is uniformly distributed over the set $\{-1, 1\}$.

— YOUR WORK —

Priority 3 — Pre-2023 Exams · Concentration Bounds

2014A · P8 (b) **MEDIUM** · 2014 Moed A exam

Closed-form bound (Markov) on probability the random MAX-3SAT assignment satisfies $< 3/4$ of clauses.

8. (14 points) Recall the MAX-3SAT problem. The input is a 3CNF formula with m clauses over n variables. The output is an assignment of truth values to the variables that maximizes the number of satisfied clauses. You may assume that the no 2 literals in the same clause belong to the same variable.
- b. (7 points) Give a bound on the probability that your algorithm satisfies less than $3/4$ of the clauses. Your bound should be in closed form.

— YOUR WORK —

Number of samples (closed form, Chernoff) to estimate π within 10% w.p. $3/4$ via unit-cube/ball sampling.

5. (10 points) Recall the randomized algorithm for computing π we discussed in class. Suppose that instead of sampling the fraction of points in a unit square that fall within a circle with radius $1/2$, we sample the fraction of points in the unit cube that fall within the a ball with radius $1/2$ (the volume of this ball is $\pi/6$). What is the number of samples required to guarantee that, with probability at least $3/4$ we get an estimate of π that is not off from the true value by more than 10%? You should give a closed form expression, but you do not need to evaluate it numerically.

— YOUR WORK —

Convert a Las Vegas algorithm to an $O(n)$ Monte Carlo algorithm w.p. $9/10$ correct via Markov's inequality.

5. (10 points) Let f be a boolean function (i.e., a function whose value is either 0 or 1). You are given a Las Vegas algorithm, $A(x)$, for determining $f(x)$. $A(x)$ always returns 1 if $f(x)=1$ and 0 if $f(x)=0$. The running time of A depends only on the length of the input ($|x|$) and on the algorithm's randomness. Let $T(n)$ be the random variable denoting the running time of A on inputs of length n . The expected running time of algorithm A on inputs of length n satisfies $E[T(n)] \leq 10n$.

Describe a randomized Monte Carlo algorithm, B , for determining $f(x)$. B should run in time $O(n)$ (where $n=|x|$), and output a correct answer with probability at least $9/10$ (the probability of a correct answer should depend only on the randomness of the algorithm, not on the input x). Prove that the algorithm you described satisfies the requirements.

Hint: Algorithm B should use algorithm A as a black box.

— YOUR WORK —

How many samples to estimate the fraction of ones within 0.01 with probability 0.9 (Chernoff bound).

4. (10 points) Recall the problem of estimating the frequency of ones in a long sequence of bits. The algorithm is given random access to the input sequence, and is required to output an estimate of the frequency of ones (number of ones divided by the length of sequence). How many samples must the algorithm make in order to guarantee that the estimate is within 0.01 from the actual frequency, with probability at least 0.9?

— YOUR WORK —

Turn Monte Carlo selection into Las Vegas: run ≤ 5 capped trials, bound failure via Markov to $1/2^5$.

- e) (8 points) Given that the expected number of operations performed by the algorithm is at most cn for some constant c , describe and analyze a randomized selection algorithm that performs at most $25cn$ operations, and returns the correct answer with probability at least $1 - \frac{1}{2^5}$.

— YOUR WORK —

Randomized — The Probabilistic Method

2 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · The Probabilistic Method

2025A · P1 (a) EASY · 2025 Moed A exam

T/F: every 3-CNF with m clauses has an assignment satisfying at most $7m/8$ clauses (averaging argument).

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(a) For every 3-CNF formula with m clauses, there exists an assignment that satisfies **at most** $7m/8$ clauses.

— YOUR WORK —

Priority 3 — Pre-2023 Exams · The Probabilistic Method

2017B · P4 (a) **MEDIUM** · 2017 Moed B exam

Prove there always exists an assignment satisfying at least $m/2$ clauses of a 3-SAT formula.

- 4 . (12 points) Recall the MAX-3-SAT problem. The input is an integer k and a set of m clauses over n boolean variables, where each clause is the disjunction (OR) of 3 literals (a literal is a variable or its negation). The goal is to decide whether there exists an assignment of the variables that satisfies at least k of the clauses.
- a) (4 points) Give a concise proof that there always exists an assignment satisfying at least $m/2$ clauses.

— YOUR WORK —

Randomized — Random Walks & Mixing

11 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · Random Walks & Mixing

2023B · P1 (d) **MEDIUM** · 2023 Moed B exam

T/F: in a regular graph with second eigenvalue $\leq 1/2$, a random walk reaches t within $C \log n$ steps w.p. $\geq 1/2n$.

Problem 1 (4×5=20 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(d) Let G be an undirected regular graph with n vertices such that all the eigenvalues in the normalized adjacency matrix of G , except the largest one, are at most $1/2$ in absolute value. There exists a constant C such that for every pair of vertices s, t in the graph, the probability that a random walk that starts from s will reach t in $C \log n$ steps is at least $\frac{1}{2n}$.

— YOUR WORK —

T/F: for a connected regular graph, the eigenspace $\{v: Av=v\}$ has dimension exactly 1 (spectral gap / connectivity).

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(d) Let G be an undirected connected regular graph with an adjacency matrix A . The dimension of the vector space $\{v \in \mathbb{R}^n : Av = v\}$ is exactly 1.

— YOUR WORK —

T/F: if $\det(A+I)=0$ for a normalized adjacency matrix then G is bipartite (-1 is an eigenvalue).

Problem 1 (4×8=32 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(d) Let G be an undirected regular connected graph with a normalized adjacency matrix A . If $\det(A+I)=0$ then G is bipartite (I is the identity matrix).

— YOUR WORK —

Prove a lazy random walk (stay $1/3$, move $2/3$) on a regular connected graph reaches t in poly steps w.p. $\geq 1/2n$.

Problem 3 (16 points)

Let G be an undirected regular connected graph and let s, t be vertices in G .

Consider the following random walk on G : starting from s , at each step the walk either stays in place with probability $1/3$, or with probability $2/3$, moves to a random neighbor (chosen uniformly and independently of all previous choices).

Prove that after a number of steps which is polynomial in n , the probability to reach t is at least $\frac{1}{2n}$.

You may use any property we proved in class regarding the adjacency matrix of G .

Priority 2 — Homework · Random Walks & Mixing

hw2025s2 · P1 (a) **EASY** · HW 2025 · Set 2

A unit eigenvector x of λ^2 has some coordinate with $x_i^2 \geq 1/n$

Part a

Let $x = (x_1, \dots, x_n)$. By the assumption $\sum_{k=1}^n x_k^2 = 1$ thus there must be i such that

— YOUR WORK —

Since x is orthogonal to the all-ones vector, some coordinate $x_j \leq 0$

Part b

As shown in class, $\mathbf{1}$ is an eigenvector corresponding to the eigenvalue 1, and thus $\langle x, \mathbf{1} \rangle = 0$ which implies that $\sum_{k=1}^n x_k = 0$. Since $x_i \geq 1/\sqrt{n}$, for the sum to be zero there must be a

— YOUR WORK —

Along a path in a connected graph there is an edge (u,v) with $|x_u - x_v| \geq 1/(n \sqrt{n})$

Part c

Since G is connected, there's a path from vertex i to vertex j . We can also assume this is a simple path, whose length is at most n . Consider the sequence of vertices on the path

— YOUR WORK —

Conclude an upper bound on λ_2 (spectral gap) from the large edge difference

Part d

The equality $\lambda_2 = \lambda_2 \langle x, x \rangle$ follows from the assumption that $\|x\|_2 = 1$. The second equality follows from basic properties of the inner product, and the second from the fact that by

— YOUR WORK —

Show a disconnected d -regular graph has $\lambda_2 = 1$ via multiplicity of eigenvalue 1

Part a

Let $S \subsetneq V$ be a non-empty connected component of G and let $x = \mathbf{1}_S$ be the indicator vector of S . Note that since $S \neq V$ and $S \neq \emptyset$, the vector x is linearly independent of $\mathbf{1}$ (since it

— YOUR WORK —

Show a bipartite d -regular graph has smallest eigenvalue $\lambda_n = -1$

Part b

If G is bipartite then up to permuting the names of the vertices, we have that

— YOUR WORK —

Priority 3 — Pre-2023 Exams · Random Walks & Mixing

2017B · P3 **MEDIUM** · 2017 Moed B exam

Using cover time $\leq 2mn$ and Markov, prove a random walk from s visits t within $8mn$ steps w.p. $\geq 3/4$.

3. (10 points) Recall that the cover time of a graph is the worst case (over choices of starting vertex) expected time for a random walk to visit all vertices in a graph. In class we said that the cover time of a graph with n vertices and m edges is at most $2mn$. Let s and t be two vertices. Prove that the probability that a random walk that starts at s visits t within $8mn$ steps is at least $3/4$.

— YOUR WORK —

Randomized — Isolation Lemma

4 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · Isolation Lemma

2023A · P1 (b) **MEDIUM** · 2023 Moed A exam

T/F: isolation lemma still holds if set weight is the product (not sum) of element weights.

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(b) The isolation lemma remains true even if we define the weight of a set to be the **product** of the weights of the elements in the set, instead of their sum.

— YOUR WORK —

Prove random edge weights in $\{1..p(n)\}$ yield a unique min-weight path for every vertex pair w.p. ≥ 0.9 (isolation lemma + union bound).

Let $G = (V, E)$ be an undirected graph and let w be a weight function that gives a non-negative integer weight for every edge in E . The weight of a path is defined to be the sum of the edges in the path.

Prove that there exists a polynomial $p(n)$ such that for every undirected n -vertex graph, if we pick for every edge a uniformly random weight from the set $\{1, \dots, p(n)\}$ (independently of all other edges), then with probability at least 0.9 the following property holds: for every pair of vertices $u, v \in V$, there's a **unique** minimum weight path between u and v .

T/F: random edge weights in $\{1, \dots, 4|E|\}$ give unique min and unique max weight cut w.p. $\geq 1/2$ (isolation lemma).

Problem 1 (4×8=32 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(c) Let $G = (V, E)$ be an undirected graph. If we assign, uniformly at random and independently, for each edge a weight from the set $\{1, 2, \dots, 4|E|\}$, then with probability at least $1/2$, there exists a unique minimum weight cut and a unique maximum weight cut (the weight of a cut is the sum of weights of edges crossing it).

— YOUR WORK —

Priority 2 — Homework · Isolation Lemma

hw2025s2 · P3 **HARD** · HW 2025 · Set 2

Find a min-weight perfect matching with exactly k red edges via the isolation lemma and symbolic determinant with weights 2^w

Problem 3

Let m denote the number of edges in G (note that $m < n^2$).

— YOUR WORK —

Algebraic — PIT & Schwartz–Zippel

12 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · PIT & Schwartz–Zippel

2023B · P1 (c) **EASY** · 2023 Moed B exam

T/F: there is a nonzero 5-variable polynomial vanishing on every integer assignment (false by Schwartz–Zippel).

Problem 1 (4×5=20 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(c) There exists a non-zero polynomial with five variables, which gets the value zero for every assignment of integers to the variables.

— YOUR WORK —

T/F: for a nonzero degree- d polynomial there exist arguments in $\{1, \dots, d+1\}$ where it is nonzero (Schwartz-Zippel).

Problem 1 (4×6=24 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(b) Let $f(x_1, \dots, x_n)$ be a non-zero polynomial over $\mathbb{Q}$. There exist $\alpha_1, \dots, \alpha_n \in \{1, 2, \dots, d+1\}$ such that $f(\alpha_1, \dots, \alpha_n) \neq 0$.

— YOUR WORK —

T/F: some nonzero polynomial of degree $< m$ vanishes on all order- m roots of unity (false by Schwartz-Zippel).

Problem 1 (4×8=32 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(a) Let $\Omega_m \subseteq \mathbb{C}$ be the set of roots of unity of order m . There exists a nonzero polynomial $f(x_1, \dots, x_n)$ such that $\deg(f) < m$ and for every $a_1, \dots, a_n \in \Omega_m$, $f(a_1, \dots, a_n) = 0$.

— YOUR WORK —

Randomized algorithm deciding whether the span of matrices $B_1..B_k$ contains an invertible matrix (Schwartz-Zippel on $\det(\sum x_i B_i)$).

In this question we'll look at vector spaces of $n \times n$ matrices.

Prove that there is a polynomial-time randomized algorithm that, given matrices $B_1, \dots, B_k \in \mathbb{R}^{n \times n}$ decides whether there exists an invertible matrix A in the subspace V spanned by $B_1, \dots, B_k$.

That is, whether there exist $x_1, \dots, x_k \in \mathbb{R}$ such that $\sum_{i=1}^k x_i B_i$ is an invertible matrix.

— YOUR WORK —

Randomized algorithm deciding determinantal equivalence of two directed graphs' symbolic adjacency matrices (Schwartz-Zippel on det difference).

Let $G = (V, E)$ an **directed** graph.

Define its symbolic adjacency matrix to be the following $n \times n$ matrix:

$$A_{i,j} = \begin{cases} x_{i,j} & \text{if } (i, j) \in E \\ 0 & \text{otherwise} \end{cases}$$

We say G_1, G_2 are **determinantly equivalent** if the determinants of their symbolic adjacency matrices are identical (as polynomials in $x_{i,j}$).

Give an efficient randomized algorithm that given a pair of directed graphs G_1, G_2 decides whether G_1, G_2 are determinantly equivalent.

The error probability should be less than 0.1.

הסימבולית של G

G הם שקולי
וּת של מטריצות
זהות (כפולינומים

ובהינתן זוג גרפים
 G_1, G_2 הם שקולי

יתם צריכה להיות

Randomized test of whether two branching programs (matrix-product polynomials) are equal, via Schwartz-Zippel.

Problem 4 (16 points)

A **branching program** over the variables $x_1, \dots, x_n$ is a set of matrices $A_1, \dots, A_m$ of size 3×3 . Every entry in every matrix contains either a variable from the set $x_1, \dots, x_n$, or a scalar from $\mathbb{Q}$.

The program computes the polynomial (over $\mathbb{Q}$) that appears in the $(1, 1)$ entry of the matrix $A_1 \cdot A_2 \cdots A_m$ obtained by multiplying $A_1, A_2, \dots, A_m$.

Show an efficient randomized algorithm that given a pair of branching programs $A_1, A_2, \dots, A_m$ and $B_1, B_2, \dots, B_m$ decides whether they compute the same polynomial over $\mathbb{Q}$.

The error probability should be less than 0.1.

— YOUR WORK —

Randomized algorithm for Matrix Completion max rank via random substitution; prove correctness, runtime, error (Schwartz-Zippel on a submatrix determinant).

In the **Matrix Completion** problem, the input is an $n \times n$ real matrix A with some of its entries marked by stars (*). The goal is to compute what is the maximal rank of a matrix that can be obtained from A by replacing the stars with real numbers (the entries marked with stars don't have to all get the same value. Each entry can be fixed independently).

For example, given the following inputs:

$$A_1 = \begin{bmatrix} 1 & * \\ * & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} * & * \\ 0 & 0 \end{bmatrix}$$

The solution for A_1 is 2 and for A_2 is 1.

Give an efficient **randomized** algorithm for the matrix completion problem. Prove its correctness, and analyze both the running time and the error probability.

Hint: think about the relation between determinants and rank.

הוא מטריצה
זיה מסומנים
מהי הדרגה
לת מ־A ע"י
קום הכוכביות
כוכבית יקבלו
בלתי תלוי).

ג 1.
יית השלמת
וריתם, נתחו
ה.
ז ודרגה.

Priority 2 — Homework · PIT & Schwartz–Zippel

hw2025s1 · P4 **MEDIUM** · HW 2025 · Set 1

Randomized equality-testing protocol with $O(\log n)$ communication via random prime modulus or polynomial fingerprint

Problem 4

Let $a \in \{0, 1\}^n$ and $b \in \{0, 1\}^n$ denote the inputs of Alice and Bob, respectively.

— YOUR WORK —

Prove the Schwartz-Zippel lemma by induction on the number of variables

Problem 1

The proof is by induction on n . For the base case, this is exactly the fact that a non-zero degree- d polynomial over a field $\mathbb{F}$ has at most d roots. For the induction step, write f as $\sum_{i=0}^d x_n^i f_i(x_1, \dots, x_{n-1})$ for some polynomials $f_0, \dots, f_d$ on $n-1$ variables with $\deg(f_i) \leq d-i$.

— YOUR WORK —

Prove a polynomial computed by an arithmetic formula of size s has degree at most s (induction on structure)

Problem 3

The proof is by induction on the structure of the formula. If the formula is a leaf, then $s = 1$. In this case, the formula is either a single variable x_i (in which case it computes a polynomial of degree 1) or a field constant c (in which case it computes a polynomial of degree 0).

— YOUR WORK —

Randomized test whether the subspace spanned by matrices $B_1 \dots B_k$ contains an invertible matrix via det + Schwartz-Zippel

Problem 4

The algorithm would work as follows: we pick uniformly at random and independently values $\alpha_1, \dots, \alpha_k$ for $x_1, \dots, x_k$ from the set $\{1, \dots, 10n\}$, and say "yes" if and only if the matrix $\sum_{i=1}^k \alpha_i B_i$ is invertible.

Clearly, if there's no invertible matrix in the subspace spanned by $B_1, \dots, B_k$, the algorithm

— YOUR WORK —

Test if two rank- n matrices share a common column basis using Cauchy-Binet and Schwartz-Zippel

Problem 5

Since by assumption $\text{rank}(U) = n$, we have that the columns of B is a basis for the column space of U iff U_B is a full rank $n \times n$ matrix, which happens iff $\det(U_B) \neq 0$. The same holds for V and V_B .

— YOUR WORK —

Algebraic — FFT & Polynomial Multiplication

10 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · FFT & Polynomial Multiplication

2025C · P1 (a) EASY · 2025 Moed C exam

T/F: if k divides n , every order- k root of unity is also an order- n root of unity.

Problem 1 (4×8=32 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(a) Let $k, n \in \mathbb{N}$ be natural numbers such that n is divisible by k . Every root of unity of order k is also a root of unity of order n .

— YOUR WORK —

$O(n \log n)$ algorithm computing the sumset $A+B$ and counts of representations via FFT polynomial multiplication.

Let A, B be subsets of $\{1, 2, \dots, n\}$. $\{1, 2, \dots,$

The **sumset** of A, B is the set $A+B$ defined $A+B$ הקבוצה
as

$$A+B = \{a+b : a \in A, b \in B\}$$

Give an algorithm that runs in time $O(n \log n)$ and given A, B computes the elements of $A+B$, and counts how many times each $c \in A+B$ is obtained as a sum of elements from A and B . O ובהינתן A, B
סופר כמה פעמים
של איברים מ- A

— YOUR WORK —

Prove the Walsh-Hadamard transform $H_n x$ can be computed in $O(N \log N)$ via an FFT-style divide-and-conquer.

Problem 2 (16 points)

שאלה 2 (16 נקודות)

Let n be a positive natural number and denote $N = 2^n$.

The **Walsh-Hadamard** matrix of order N is an $N \times N$ matrix H_n defined inductively as follows:

יהי n מספר טבעי חיובי ונסמן $N = 2^n$.

מטריצת **וולש-הדמרד** מסדר N היא מטריצה H_n ממיד $N \times N$ המוגדרת באינדוקציה באופן הבא:

$$H_1 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H_n = \begin{bmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -H_{n-1} \end{bmatrix} \quad \forall n \geq 2$$

Prove that there is an algorithm that, on input $x \in \mathbb{R}^N$, computes the vector $H_n x$ in time $O(N \log N)$.

הוכיחו כי קיים אלגוריתם אשר בהינתן קלט $x \in \mathbb{R}^N$ מחשב את הוקטור $H_n x$ בזמן $O(N \log N)$.

— YOUR WORK —

T/F: coefficients of $f(x)^n$ computable in $O(n^2 \log n)$ ops via repeated squaring + FFT multiplication.

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(b) There exists an algorithm that given a polynomial $f(x)$ of degree n , computes the coefficients of the polynomials $f(x)^n$ with $O(n^2 \log n)$ arithmetic operations (you may assume that n is a power of 2).

— YOUR WORK —

Given f, g with $f=g \cdot h$, find coefficients of h in $O(n \log n)$ via evaluate-at-roots (FFT), divide, interpolate.

Problem 4 (16 points)

Let $f(x), g(x)$ be polynomials such that $\deg(g) \leq \deg(f) < n$ and $m \leq n$.

Assume further that there exists a polynomial h of degree $\deg f - \deg g$ such that $g \cdot h = f$, and $g(\omega^i) \neq 0$ for every $0 \leq i \leq n-1$ (where ω is a primitive root of unity of order n).

Give an algorithm that given f, g as above, finds the coefficients of h in time $O(n \log n)$.

— YOUR WORK —

Priority 2 — Homework · FFT & Polynomial Multiplication

hw2025s3 · P2 (a) **EASY** · HW 2025 · Set 3

Karatsuba: compute the cross term $x_0y_1+x_1y_0$ with a single extra multiplication

Part a

We have that $(x_0+x_1)(y_0+y_1) = x_0y_0+x_0y_1+x_1y_0+x_1y_1$. Thus, we can compute the elements we need by first computing x_0y_0 , x_1y_1 then computing $(x_0+x_1)(y_0+y_1) - x_0y_0 - x_1y_1$. Note

— YOUR WORK —

Solve the Karatsuba recurrence $T(n)=3T(n/2)+O(n) = O(n^{\log_2(3)})$

Part b

Let $T(n)$ denote the running time of the algorithm when multiplying two n -bit integers. The previous item gives a way to compute the multiplication recursively, using 3 multiplications

— YOUR WORK —

Multiply two bivariate polynomials by Kronecker substitution $y=x^{(2n+1)}$, reducing to one univariate FFT product

Problem 1

By collecting monomials with similar y -degree, write $f(x, y) = \sum_{j=0}^d f_j(x) y^j$ and $g(x, y) =$

— YOUR WORK —

Count the ways to write $k=i+j$ with i in A , j in B (sumset) via polynomial multiplication with FFT

Problem 3

We associate with A the polynomial $f_A = \sum_{i \in A} x^i$ and with B the polynomial $f_B = \sum_{i \in B} x^i$.

— YOUR WORK —

Priority 3 — Pre-2023 Exams · FFT & Polynomial Multiplication

2014A · P6 **MEDIUM** · 2014 Moed A exam

Prove two n -bit integers can be multiplied in $O(n^{\log_2 3})$ (Karatsuba divide-and-conquer).

6. (8 points) Prove that two n -bit integers can be multiplied in $O(n^{\log_2 3})$ time. You may use the following basic operations: a) addition of two k -bit numbers, which takes $O(k)$ time, and b) multiplying a k -bit number by 2^m (shift), which takes $O(k+m)$ time.

Let A, C be two n -bit numbers. Write $A = a2^{n/2} + b$, and $C = c2^{n/2} + d$. Now $A \cdot C =$

— YOUR WORK —

Algebraic — Matrix-Multiplication Applications

19 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · Matrix-Multiplication Applications

2023A · P1 (d) **EASY** · 2023 Moed A exam

T/F: two 4×4 matrices can be multiplied with at most 49 multiplications (Strassen, one recursion level).

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(d) There exists an algorithm for multiplying two 4×4 matrices that performs at most 49 multiplications.

— YOUR WORK —

T/F: a symmetric matrix with $\det \geq 0$ is PSD (false; counterexample -I).

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(b) Let A be a symmetric matrix. If $\det(A) \geq 0$, then A is PSD.

— YOUR WORK —

T/F: poly-time algorithm exists to find an orthonormal basis of a span (Gram-Schmidt).

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(c) There exists a polynomial time algorithm that given $v_1, \dots, v_k \in \mathbb{R}^n$ finds an orthonormal basis for the subspace spanned by $v_1, \dots, v_k$.

— YOUR WORK —

Compute the boolean matrix product $A \circ B$ in time $o(n^3)$ by casting it to integer matrix multiplication.

Problem 2 (14 points)

שאלה 2 (14 נקודות)

Let A, B be $n \times n$ matrices whose entries are in $\{0, 1\}$. The **boolean multiplication** of A and B is an $n \times n$ matrix defined as

תהיינה A, B מטריצות בגודל $n \times n$ שכל איבריהן בקבוצה $\{0, 1\}$. הכפל הבוליאני של A ו- B היא מטריצה $n \times n$ המוגדרת ע"י

$$(A \circ B)_{i,j} = \bigvee_{k=1}^n (A_{i,k} \wedge B_{k,j}), \quad 1 \leq i, j \leq n$$

Give an algorithm that given A, B computes the matrix $A \circ B$ in time which is asymptotically smaller than n^3 .

תנו אלגוריתם אשר בהינתן A, B מחשב את המטריצה $A \circ B$ בזמן קטן אסימפטוטית מ- n^3 .

— YOUR WORK —

T/F: computing n products $Av_1 \dots Av_n$ requires time n^3 (false; fast matrix multiplication gives $O(n^\omega)$).

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(c) Let A be an $n \times n$ matrix. If $v_1, \dots, v_n$ are linearly independent vectors, computing the n products $Av_1, \dots, Av_n$ requires time n^3 .

— YOUR WORK —

T/F: an $n \times n$ by $n \times n^2$ matrix product can be done in fewer than n^4 operations (block decomposition, $n^{\{w+1\}}$).

Problem 1 (4×8=32 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(b) There exists an algorithm for multiplying an $n \times n$ matrix by an $n \times n^2$ matrix that performs less than n^4 arithmetic operations.

— YOUR WORK —

Compute transitive closure of a digraph in time $O(n^3)$ via $(A+I)^n$ by fast matrix mult and repeated squaring.

Problem 5 (16 points)

Let $G = (V, E)$ be a directed graph.

The **transitive closure** of G is the graph $G^* = (V, E^*)$ such that for every $u, v \in V$, $(u, v) \in E^*$ if there's a path from u to v in G . Give an algorithm that, given a graph G , computes G^* , and whose running time is asymptotically smaller than n^3 . Note that this should hold for the total running time, and not for the parallel time.

— YOUR WORK —

T/F: the n -th elementary symmetric polynomial in eigenvalues is efficiently computable via characteristic-poly coefficients.

Problem 1
(4×6=24 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(d) Let $A \in \mathbb{C}^{2n \times 2n}$ a matrix with eigenvalues $\lambda_1, \dots, \lambda_{2n}$. There is an efficient algorithm that given A computes:

ד. תהי $A \in \mathbb{C}^{2n \times 2n}$ מטריצה עם ערכים עצמיים $\lambda_1, \dots, \lambda_{2n}$. קיים אלגוריתם יעיל שבהינתן A מחשב את ערך הביטוי:

$$\sum_{\substack{S \subseteq \{1, \dots, 2n\} \\ |S|=n}} \prod_{i \in S} \lambda_i$$

— YOUR WORK —

Compute the diameter of a strongly connected digraph in time $o(n^3)$ via matrix powers and binary search.

Problem 4 (14 points)

Let $G = (V, E)$ be a strongly connected **directed** graph. For a pair of vertices $u, v \in V$, let $d(u, v)$ denote the distance between u and v , that is, the length (in edges) of a shortest path from u to v .

The **diameter** of G is defined as $\max_{u \neq v \in V} d(u, v)$.

Give an algorithm, that runs in time asymptotically smaller than n^3 , and given a strongly connected directed graph G , computes its diameter.

— YOUR WORK —

T/F: the min multiplications for matrix mult satisfy $a_{k^2} \leq (a_k)^2$ (recursive block multiplication).

Problem 1 (4×8=32 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(b) Let α_k be the minimal number of multiplications needed for computing $k \times k$ matrix multiplication. Then, it holds that $\alpha_{k^2} \leq (\alpha_k)^2$.

— YOUR WORK —

Priority 2 — Homework · Matrix-Multiplication Applications

hw2025s2 · P4 **MEDIUM** · HW 2025 · Set 2

Count triangles in a graph via the trace of A^3 using fast matrix multiplication

Problem 4

Let A be the $n \times n$ adjacency matrix of G : i.e., $A_{i,j} = 1$ if there's an edge between the

— YOUR WORK —

Compute transitive closure via $(A+I)^n$ using fast matrix multiplication and repeated squaring

Problem 5

Let A be the adjacency matrix of G . As we've seen in class, $(A^k)_{i,j}$ counts the number of

— YOUR WORK —

Show $\text{rank}(A) = \text{rank}(A^T A)$ by proving the two matrices have equal kernels

Part a

Let $B = A^T A$. We will show that for every $v \in \mathbb{R}^m$, $Av = 0$ if and only if $Bv = 0$. This implies that $\ker(A) = \ker(B)$ and thus $\text{rank}(A) = \text{rank}(B)$.

— YOUR WORK —

Show $\text{rank}(B) = m$ minus the number of zero eigenvalues for symmetric B

Part b

As in part (a), we will use the fact that if $\dim \ker(B) = k$ then $\text{rank}(B) = m - k$. The rank of the kernel is exactly the geometric multiplicity of the eigenvalue 0 of B . However, since B

— YOUR WORK —

Parallel algorithm computing $\text{rank}(A)$ via the characteristic polynomial of $A^T A$ (smallest nonzero coefficient)

Part c

By part (b), 0 is a root of p with multiplicity k , which implies that x^k divides $p(x)$. Further, x^k is the largest power of x dividing $p(x)$ (as otherwise the multiplicity of 0 would have been

— YOUR WORK —

Show a PSD matrix ($v^T A v \geq 0$) has all eigenvalues ≥ 0

Part a

Let $u \neq 0$ be an eigenvector corresponding to an eigenvalue λ . By the assumption, $u^T A u \geq 0$.

— YOUR WORK —

Show a PSD matrix A can be written as $A = B^T B$ via diagonalization

Part b

Since A is symmetric, it can diagonalized as $A = PDP^{-1}$ where P is a matrix whose

— YOUR WORK —

Show $A = B^T B$ implies A is positive semidefinite

Part c

Suppose that $A = B^T B$ for some B . Then for every $v \in \mathbb{R}^n$,

— YOUR WORK —

Priority 3 — Pre-2023 Exams · Matrix-Multiplication Applications

2018A · P1 (a) **EASY** · 2018 Moed A exam

Write and solve Strassen's recurrence $T(n)=7T(n/2)+\Theta(n^2)$ giving $\Theta(n^{\log_2(7)})$.

1. (5x6 points)

- a) Recall that Strassen's algorithm for matrix multiplication takes as input two n -by- n matrices, and computes their product by recursively computing 7 products of $n/2$ -by- $n/2$ matrices and 10 additions of $n/2$ -by- $n/2$ matrices. Write down and solve the recurrence for the running time of Strassen's algorithm.

— YOUR WORK —

Algebraic — Algebraic Bipartite Matching

1 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · Algebraic Bipartite Matching

2024C · P1 (d) **MEDIUM** · 2024 Moed C exam

T/F: parity of the number of perfect matchings in a bipartite graph is efficiently decidable via $\det(A) \bmod 2$.

Problem 1 (4×6=24 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(d) There exists an efficient algorithm that decides, given a bipartite graph G , whether the number of perfect matchings in G is odd.

— YOUR WORK —

Linear Programming — LP Basics & Geometry

27 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · LP Basics & Geometry

2023A · P1 (a) **EASY** · 2023 Moed A exam

T/F: simplex can move to a different vertex whose objective value equals the previous vertex's (degeneracy).

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(a) In the simplex algorithm as presented in class, it is possible that the algorithm moves to a different vertex, and the value of the objective function in the new vertex equals its value in the previous vertex.

— YOUR WORK —

T/F: each pivoting step of the simplex algorithm runs in polynomial time in the input length.

Problem 1 (4×5=20 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(b) Each pivoting step in the simplex algorithm takes polynomial time in the input length.

— YOUR WORK —

T/F: there is an efficient algorithm to decide whether a given point y is a vertex of a standard-form polytope.

Problem 1 (4×6=24 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(b) There exists an efficient algorithm that, given an LP in standard form and a point y , decides if y is a vertex of the polytope that corresponds to the LP.

— YOUR WORK —

T/F: a standard-form polytope in $\mathbb{R}^n$ has at most 2^n vertices (one per subset of columns with full rank).

Problem 1 (4×6=24 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(a) Let $P = \{x : Ax = b, x \geq 0\} \subseteq \mathbb{R}^n$ be a polytope in standard form. The number of vertices of P is at most 2^n .

— YOUR WORK —

T/F: x is optimal for an LP iff x is a vertex (false; interior/edge optima exist).

Problem 1 (4×8=32 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(d) Let (P) be a linear program. Then x is an optimal solution for (P) iff x is a vertex.

— YOUR WORK —

Relax the maximum-matching IP to an LP and write it in standard form (with slack variables).

Note: in this question, the graphs are not necessarily bipartite. זו הגרפים אינם בהכרח

As seen in class, given a graph $G = (V, E)$, the **Maximum Matching** problem can be expressed as the following integer program: בהינתן גרף $G = (V, E)$, זימלי ניתנת לכתיבה בתור באופן הבא:

$$\begin{array}{ll}
 \text{maximize} & \sum_{(u,v) \in E} x_{u,v} \\
 \text{subject to} & \sum_{u:(u,v) \in E} x_{u,v} \leq 1 \quad \text{for all } v \in V \\
 & x_{u,v} \geq 0 \quad \text{for all } (u,v) \in E \\
 & x_{u,v} \text{ integer} \quad \text{for all } (u,v) \in E
 \end{array}$$

(a) We'll relax the integer program by removing the last constraint. Write the resulting linear program in standard form (the form suitable for running the simplex algorithm).

— YOUR WORK —

T/F: if the reduced-cost vector is strictly positive at a BFS, that BFS is the unique optimum.

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(d) Let x be a basic feasible solution with a basis B for a linear program (P) in standard form. Let $\tilde{c}_N$ be the reduced cost vector corresponding to z . If in the vector $\tilde{c}_N$ all entries are strictly positive, then z is the unique optimal solution.

— YOUR WORK —

Prove that if the reduced cost vector is nonnegative then the basic feasible solution is optimal.

Problem 3 (16 points)

Let x be a basic feasible solution with a basis B to a linear program in standard form. Let $\tilde{c}_N$ be the reduced cost vector associated with x . Prove that if $\tilde{c}_N \geq 0$ then x is an optimal solution.

— YOUR WORK —

Priority 2 — Homework · LP Basics & Geometry

hw2025s3 · P4 **EASY** · HW 2025 · Set 3

Formulate a transportation / shipping-cost minimization problem as a linear program and solve it

Problem 4

For each field F and city C we will have a variable $x_{F,C}$ corresponding to how many tons

— YOUR WORK —

If $\{Cx \leq d\}$ is feasible, a ball of radius $2^{-2L'}$ around a feasible point lies inside $\{Cx \leq d + 2^{-L'} \mathbf{1}\}$

1. In this exercise we will prove the claim we made in class, that when we need to decide the feasibility problem for a polyhedron $P = \{x : Cx \leq d\}$ we can assume without loss of generality a lower bound on its volume.

Recall that in class we proved the volume upper bound: that is, for every vertex x of P and for every coordinate j , $|x_j| \leq 2^L$ where L an integer whose size is at most a polynomial function of the input.

Let $\mathbf{1}$ denote the all ones vector. In the following two parts, L' stands for some integer whose size is at most a polynomial function of the input. You don't need to compute it exactly, you just need to show that one can pick L' with that property such that the statements hold.

- (b) Prove that if $Cx \leq d$ is feasible, there's a feasible point $\hat{x}$ such that the ball of radius $2^{-2L'}$ around $\hat{x}$, that is, the set $\{x : \|x - \hat{x}\|_2 \leq 2^{-2L'}\}$, is contained in $\{x : Cx \leq d + 2^{-L'} \mathbf{1}\}$.

— YOUR WORK —

Bland's-rule termination: establish sign facts about the pivot column ($A_{ts} > 0$, $s < t$, reduced costs)

Part a

- i. The fact that $\bar{A}_{t,s} > 0$ follows from the fact that s enters and t leaves the basis: the

— YOUR WORK —

Bland's-rule termination: construct direction d and prove its properties ($d_j \geq 0$, $Ad=0$)

Part b

- i. By definition, $d_t = -\bar{A}_{t,s}$, and as we argued above, $\bar{A}_{t,s} > 0$. Further, $d_s = 1$ by

— YOUR WORK —

Bland's-rule termination: reduced-cost sign facts at the new basis B'

Part c

- i. Since t is entering the new basis, $\tilde{c}'_t < 0$ by the description of the simplex algorithm.

— YOUR WORK —

Bland's-rule termination: show cost difference $f = v^T A$ so that $f \cdot d = 0$

Part d

We have that

— YOUR WORK —

Bland's-rule termination: sign analysis of the coordinates of f

Part e

We use the properties of $\tilde{c}$ and $\tilde{c}'$ we proved above. $\tilde{f}_t = \tilde{c}'_t - \tilde{c}_t < 0$ since $\tilde{c}'_t < 0$ and $\tilde{c}_t = 0$.
 $\tilde{f}_s = \tilde{c}'_s - \tilde{c}_s > 0$ since $\tilde{c}'_s \geq 0$ and $\tilde{c}_s < 0$.

— YOUR WORK —

Bland's-rule termination: derive the contradiction $f \cdot d > 0$, proving the simplex cannot cycle

Part f

We now write:

— YOUR WORK —

Convert a given LP with equalities and sign constraints into standard form.

d. (5 points) Write the following LP in standard form

$$\max 2x_1 + 5x_2 - 3x_3 \text{ s.t.}$$

$$x_1 + 2x_2 = 3$$

$$x_2 - 3x_3 \leq 10$$

$$x_1 \leq 0$$

$$x_2 \geq 0$$

$$x_3 \leq 0$$

— YOUR WORK —

Add slack variables and put a given LP into standard form for the simplex algorithm.

10. (15 points) Consider the following LP:

$$\max 2x_1 + 5x_2 + 2x_3 \text{ s.t.}$$

$$x_1 + 2x_2 \leq 10$$

$$2x_1 + x_2 - x_3 \geq 14$$

$$x_i \geq 0 \forall i$$

- a. (3 points) Add non-negative slack variables and write down the system in the standard form required for running the simplex algorithm.

— YOUR WORK —

Find the basic-feasible-solution vertex associated with a given basis $\{x_1, s_2\}$.

10. (15 points) Consider the following LP:

$$\max 2x_1 + 5x_2 + 2x_3 \text{ s.t.}$$

$$x_1 + 2x_2 \leq 10$$

$$2x_1 + x_2 - x_3 \geq 14$$

$$x_i \geq 0 \forall i$$

b. (3 points) What is the vertex associated with the basis x_1, s_2 ?

— YOUR WORK —

Determine which non-basic variable first enters the basis in a simplex pivot.

10. (15 points) Consider the following LP:

$$\max 2x_1 + 5x_2 + 2x_3 \text{ s.t.}$$

$$x_1 + 2x_2 \leq 10$$

$$2x_1 + x_2 - x_3 \geq 14$$

$$x_i \geq 0 \forall i$$

c. (3 points) Consider an execution of the simplex algorithm starting with the basis x_1, s_2 . Which non basic variable is the first to enter the basis?

— YOUR WORK —

Determine which basic variable leaves the basis via the min-ratio rule in a simplex pivot.

10. (15 points) Consider the following LP:

$$\max 2x_1 + 5x_2 + 2x_3 \text{ s.t.}$$

$$x_1 + 2x_2 \leq 10$$

$$2x_1 + x_2 - x_3 \geq 14$$

$$x_i \geq 0 \forall i$$

d. (3 points) Which basic variable leaves the basis at that time?

— YOUR WORK —

Compute the objective value after the first simplex pivot.

10. (15 points) Consider the following LP:

$$\max 2x_1 + 5x_2 + 2x_3 \text{ s.t.}$$

$$x_1 + 2x_2 \leq 10$$

$$2x_1 + x_2 - x_3 \geq 14$$

$$x_i \geq 0 \forall i$$

e. (3 points) What is the value of the objective function after the first pivot?

— YOUR WORK —

Add slack variables and perform the first simplex pivot, writing objective and basics as non-basic functions.

7. (24 points) Consider the following LP:

$$\begin{aligned} \max \quad & 2x_1 + 4x_2 + 3x_3 + x_4 \quad s. t. \\ & 3x_1 + x_2 + x_3 + 4x_4 \leq 12 \\ & x_1 - 3x_2 + 2x_3 + 3x_4 \leq 7 \\ & 2x_1 + x_2 + 3x_3 - x_4 \leq 10 \\ & x \geq 0 \end{aligned}$$

- a. (6 points) Add slack variables and perform the first pivot of the Simplex algorithm. Write down the objective function and basic variables as a function of the non-basic variables after the first pivot.

— YOUR WORK —

Model $\min |2x+y-3| + |10x+y+5|$ as a linear program using auxiliary variables.

- d. (6 points) Write down the following minimization problem with variables x, y as a linear program: $\min |2x + y - 3| + |10x + y + 5|$

— YOUR WORK —

Add slack variables and write a given LP in the standard form for the simplex algorithm.

d. (5 points) Consider the LP:

$$\max 5x_1 - 7x_2 + 2x_3 \text{ s.t.}$$

$$2x_1 - x_2 \geq -5$$

$$3x_1 - x_2 + 2x_3 \leq 9$$

$$x_i \geq 0 \forall i$$

Add slack variables and write down the system in the form required for running the simplex algorithm.

— YOUR WORK —

Find the vertex associated with a given simplex basis $\{x_1, s_1\}$.

- e. (5 points) What is the vertex associated with the basis x_1, s_1 ?
The non-basic variables are x_2, x_3, s_2 , so

— YOUR WORK —

Express $\min |2x-10|$ subject to $4x-5 \geq 0$ as a linear program (auxiliary variable z).

d) (5 points) Write down the following minimization problem as an LP.

$$\begin{array}{ll} \min |2x - 10| & \text{s.t.} \\ 4x - 5 \geq 0 \end{array}$$

— YOUR WORK —

Linear Programming — LP Duality & Farkas

29 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · LP Duality & Farkas

2023B · P1 (a) **EASY** · 2023 Moed B exam

T/F: if primal polyhedron is bounded and non-empty, primal and dual optima coincide (strong duality).

Problem 1 (4×5=20 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(a) In a linear program, if the polyhedron of the primal problem is bounded and non-empty, the optimum of the primal and the dual are the same.

— YOUR WORK —

T/F: if primal and dual are both feasible, the primal optimum can be strictly greater than the dual optimum (strong duality).

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(a) Let (P) a linear program in standard form, and let (D) be its dual.

It is possible that both programs are feasible, and the optimum of (P) is strictly greater than the optimum of (D).

— YOUR WORK —

Write and simplify the dual LP of the fractional-matching LP from part (a).

Note: in this question, the graphs are not necessarily bipartite. זו הגרפים אינם בהכרח

As seen in class, given a graph $G = (V, E)$, the **Maximum Matching** problem can be expressed as the following integer program: בהינתן גרף $G = (V, E)$, זימלי ניתנת לכתיבה בתור באופן הבא:

$$\begin{array}{ll}
 \text{maximize} & \sum_{(u,v) \in E} x_{u,v} \\
 \text{subject to} & \sum_{u:(u,v) \in E} x_{u,v} \leq 1 \quad \text{for all } v \in V \\
 & x_{u,v} \geq 0 \quad \text{for all } (u,v) \in E \\
 & x_{u,v} \text{ integer} \quad \text{for all } (u,v) \in E
 \end{array}$$

(b) Write the dual LP for the LP you got in item (a). Try to simplify your answer as much as possible.

— YOUR WORK —

Identify that adding integrality to the dual LP gives Vertex Cover (dual of matching; Konig's theorem).

Note: in this question, the graphs are not necessarily bipartite. זו הגרפים אינם בהכרח

As seen in class, given a graph $G = (V, E)$, the **Maximum Matching** problem can be expressed as the following integer program: בהינתן גרף $G = (V, E)$, זימלי ניתנת לכתיבה בתור באופן הבא:

$$\begin{array}{ll}
 \text{maximize} & \sum_{(u,v) \in E} x_{u,v} \\
 \text{subject to} & \sum_{u:(u,v) \in E} x_{u,v} \leq 1 \quad \text{for all } v \in V \\
 & x_{u,v} \geq 0 \quad \text{for all } (u,v) \in E \\
 & x_{u,v} \text{ integer} \quad \text{for all } (u,v) \in E
 \end{array}$$

(c) When you add integrality constraint to the LP you got in item (b), it corresponds to a well known graph problem. Explain which problem it is (if you don't recall the name, describe the problem briefly and clearly).

— YOUR WORK —

State and prove the weak duality theorem for LPs in standard form.

Problem 2 (12 points)

State and prove the **weak duality theorem** for linear programs in standard form.

— YOUR WORK —

Prove a case of strong duality via Farkas: if (P) is infeasible and (D) feasible then (D) is unbounded.

Problem 6 (16 points)**שאלה 6** (16 נקודות)

Prove the following case of the strong duality theorem. Given the pair of linear programs:

הוכיחו את המקרה הבא של משפט הדואליות החזק. בהינתן זוג התוכניות הלינאריות:

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b, x \geq 0 \\ (P) \end{array}$$

$$\begin{array}{ll} \text{maximize} & b^T y \\ \text{subject to} & A^T y \leq c \\ (D) \end{array}$$

Prove that if (P) is infeasible and (D) is feasible, then (D) is unbounded.

הוכיחו כי אם (P) אינה פיזיבילית ו-(D) פיזיבילית, אז (D) אינה חסומה.

— YOUR WORK —

Compute the dual of an LP with symmetric matrix A ($\min c^T x$ s.t. $Ax \geq c$, $x \geq 0$).

Let A be a symmetric $n \times n$ matrix. Consider the following LP:

$$\begin{aligned} \text{(P)} \quad & \text{minimize} && c^T x \\ & \text{subject to} && Ax \geq c \\ & && x \geq 0 \end{aligned}$$

(a) Compute the dual of (P).

Solution: We will first put (P) in standard

— YOUR WORK —

Show that x^* with $Ax^*=c$ and $x^*\geq 0$ is optimal for the LP (strong duality / self-dual argument).

Let A be a symmetric $n \times n$ matrix. Consider the following LP:

$$\begin{aligned} \text{(P)} \quad & \text{minimize} && c^T x \\ & \text{subject to} && Ax \geq c \\ & && x \geq 0 \end{aligned}$$

(b) Show that if x^* satisfies $Ax^* = c$ and $x^* \geq 0$, then x^* is an optimal solution of (P).

— YOUR WORK —

Show the stated LP is the dual of the set-cover LP relaxation by reducing to standard form.

Problem 2 (10+10=20 points)

שאלה 2 (20=10+10 נקודות)

In the set cover problem, the input is a family of subsets $S_1, \dots, S_m \subseteq \{1, 2, \dots, n\}$ where every set has a weight c_i . Every $j \in \{1, 2, \dots, n\}$ appears in at least one of the sets. The goal is to find a family of sets whose union is $\{1, 2, \dots, n\}$, where the total weight is minimal (the weight of a family is the sum of weights of sets in it).

Recall the integer LP that corresponds to this problem:

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^m c_i x_i \\ \text{subject to} & \sum_{i: S_i \ni j} x_i \geq 1 \quad \text{for every } 1 \leq j \leq n \\ & x_i \in \{0, 1\} \quad \text{for every } 1 \leq i \leq m \quad (1) \end{array}$$

(a) We will relax the IP by replacing the constraints (1) by $0 \leq x_i$. Show that the following LP is the dual of the relaxation:

$$\begin{array}{ll} \text{(D) maximize} & \sum_{j=1}^n y_j \\ \text{subject to} & \sum_{j: j \in S_i} y_j \leq c_i \quad \text{for every } 1 \leq i \leq m \quad (2) \\ & y_j \geq 0 \quad \text{for every } 1 \leq j \leq n \end{array}$$

בבעיית הכיסוי בקבוצות הקלט הוא משפחה של תתי קבוצות $S_1, \dots, S_m \subseteq \{1, 2, \dots, n\}$ כאשר לכל קבוצה משקל c_i . כל איבר $j \in \{1, 2, \dots, n\}$ מופיע בלפחות אחת מהקבוצות. המטרה היא למצוא משפחה של קבוצות שאיחודה הוא $\{1, 2, \dots, n\}$ עם משקל כולל מינימלי (משקל של משפחה הוא סכום משקלי הקבוצות שבה).

ניזכר בתכנית הלינארית בשלמים המתאימה לבעייה:

א. נבצע רלקסציה לתוכנית ונחליף את קבוצת האילוצים (1) באילוצים $0 \leq x_i$. הראו כי התכנית הלינארית הבאה היא התכנית הדואלית לרלקסציה:

— YOUR WORK —

Priority 2 — Homework · LP Duality & Farkas

hw2025s4 · P1 **MEDIUM** · HW 2025 · Set 4

Derive a standard-form variant of Farkas' lemma (exists $y \geq 0$, $y^T A \geq 0$, $y^T b < 0$) from the class version

Problem 1

The version of Farkas' lemma we proved in class stated that if $\{Cx \leq d\}$ is unsolvable,

— YOUR WORK —

Primal infeasible plus dual feasible implies the dual is unbounded (via Farkas)

Part a

Let y be a feasible solution for (D). Since (P) is infeasible, by the version Farkas Lemma from Problem 1, there's y' such that $(y')^T A \geq 0$ and $(y')^T b < 0$.

— YOUR WORK —

Dual infeasible plus primal feasible implies the primal is unbounded (via Farkas)

Part b

The proof is nearly identical. Let x be a feasible solution for (P). Since (D) is infeasible, by the version of Farkas' Lemma from class, there's $x' \geq 0$ such that $(x')^T A^T = 0$ and

— YOUR WORK —

Farkas II: if $\{Cx \leq d\}$ is infeasible then the perturbed $\{Cx \leq d + 2^{-L'} \mathbf{1}\}$ is still infeasible

1. In this exercise we will prove the claim we made in class, that when we need to decide the feasibility problem for a polyhedron $P = \{x : Cx \leq d\}$ we can assume without loss of generality a lower bound on its volume.

Recall that in class we proved the volume upper bound: that is, for every vertex x of P and for every coordinate j , $|x_j| \leq 2^L$ where L an integer whose size is at most a polynomial function of the input.

Let $\mathbf{1}$ denote the all ones vector. In the following two parts, L' stands for some integer whose size is at most a polynomial function of the input. You don't need to compute it exactly, you just need to show that one can pick L' with that property such that the statements hold.

- (a) Prove that if $Cx \leq d$ is infeasible, then so is $Cx \leq d + 2^{-L'} \mathbf{1}$ (Hint: Farkas' Lemma)

— YOUR WORK —

Priority 3 — Pre-2023 Exams · LP Duality & Farkas

2017A · P6 (b) **EASY** · 2017 Moed A exam

Write the dual of the LP relaxation of the minimum hitting set program.

- b. (5 points) Write down the dual of the LP relaxation of the program you wrote in (a)

— YOUR WORK —

Describe the combinatorial problem given by the ILP version of the dual (max disjoint sets / set packing).

- c. (5 points) Describe in a few words the problem expressed by the BIP version of the LP in (b)

— YOUR WORK —

Write the dual of the vertex-cover LP relaxation.

8. Let V be a finite set. Let E be a set of unordered pairs of elements in V .
- c. Write DLP1, the dual of LP1.

— YOUR WORK —

Interpret the dual ILP of vertex cover as the maximum matching problem.

8. Let V be a finite set. Let E be a set of unordered pairs of elements in V .
- d. Interpret the meaning of the integer program corresponding to DLP1 when V and E are the vertex and edge set of a graph. What is the name of this problem?

— YOUR WORK —

Write the dual linear program of a given maximization LP.

b. (6 points) Write down the dual linear program of the LP in (a)

— YOUR WORK —

Prove the dual optimum is at least 40 using a feasible primal solution and weak/strong duality.

- c. (6 points) Prove that the optimum objective value of the dual problem you wrote in (b) is at least 40.

In part (a) we obtained a feasible solution to the maximization LP whose value is 40.

— YOUR WORK —

Write the dual of the independent-set LP relaxation.

e. (4 points) Write down the dual of the LP in (b).

— YOUR WORK —

Interpret the dual ILP as a graph problem (minimum edge cover).

- f. (3 points) Consider the LP in (e) as an ILP. What is the graph problem described by this ILP? (you may either describe the problem in a few words, or just name it).

— YOUR WORK —

Write the dual of the LP relaxation of the maximum-weight matching program.

- b. (5 points) Write down the dual of the LP relaxation of the program you wrote in (a)

— YOUR WORK —

Describe the problem expressed by the matching dual LP (weighted vertex cover).

c. (5 points) Describe in a few words the problem expressed by the LP in (b)

— YOUR WORK —

Write the dual of the knapsack LP relaxation.

d) (5 points) Write down the dual of the relaxation from (c).

— YOUR WORK —

Write the dual of the matching LP relaxation and state the dual variables.

c) (5 points) Write down the dual of the LP from item b. Clearly state what are the variables.

— YOUR WORK —

For a given instance, write the primal in matrix/vector form and its dual BIP with variables $y_1..y_4$.

שאלה 2 (18 נק')

נתונה קבוצה של n סטודנטים $A = \{a_1, a_2, \dots, a_n\}$ ו- m קורסים. לכל $1 \leq j \leq m$, בקורס ה- j משתתפת קבוצת הסטודנטים $U_j \subseteq A$. בנוסף נתון לכל קורס פרמטר c_j . עליכם לבחור קבוצה גדולה ביותר של סטודנטים תחת המגבלה שלכל קורס $1 \leq j \leq m$ יהיו בקבוצת הסטודנטים הנבחרת לכל היותר c_j נציגים מתוך U_j . יתכן שסטודנט שנבחר משתתף במספר קורסים, ואז הוא נספר כנציג עבור כל אחד מהם.

2.2 (6 נק') תארו בכתב וקטורי ומטריציוני עבור הקלט הבא את הבעיה, וכן את הבעיה הדואלית המתאימה, גם היא כ-BIP. סמנו את המשתנים של הבעיה הדואלית ב- $y_1 \dots y_4$.

$n=5, m=4, A=\{a_1, a_2, a_3, a_4, a_5\}, U_1=\{a_1, a_3\}, U_2=\{a_2, a_3, a_4, a_5\},$
 $U_3=\{a_1, a_3, a_4, a_5\}, U_4=\{a_2, a_5\}. c_1=1, c_2=2, c_3=2, c_4=1$

— YOUR WORK —

Describe the general dual BIP (course-selection / covering problem) and interpret its variables, constraints, and objective.

שאלה 2 (18 נק')

נתונה קבוצה של n סטודנטים $A = \{a_1, a_2, \dots, a_n\}$ ו- m קורסים. לכל $1 \leq j \leq m$, בקורס ה- j משתתפת קבוצת הסטודנטים $U_j \subseteq A$. בנוסף נתון לכל קורס פרמטר c_j . עליכם לבחור קבוצה גדולה ביותר של סטודנטים תחת המגבלה שלכל קורס $1 \leq j \leq m$ יהיו בקבוצת הסטודנטים הנבחרת לכל היותר c_j נציגים מתוך U_j . יתכן שסטודנט שנבחר משתתף במספר קורסים, ואז הוא נספר כנציג עבור כל אחד מהם.

2.3 (4 נק') תארו את הבעיה הדואלית של הבעיה הפרימאלית (הכללית, מסעיף 2.1) כבעיית BIP. הסבירו במילים מהם המשתנים מהם האילוצים ומהי משמעות פונק' המטרה.

— YOUR WORK —

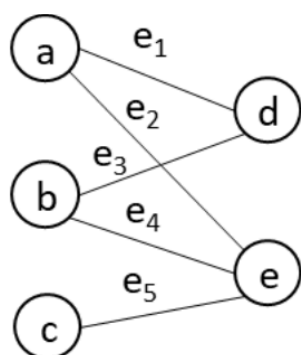
Write the semi-matching BIP in matrix/vector form for the given graph and its dual BIP.

שאלה 1 (18 נק')

יהי $G=(V,E)$ גרף דו-צדדי לא מכוון, כלומר $V=V_1\cup V_2$, $E\subseteq V_1\times V_2$. קבוצת קשתות $E'\subseteq E$ מהווה **semi-matching** אם בגרף $G=(V,E')$ דרגת כל צומת ב- V_1 היא 0 או 1, ודרגת כל צומת ב- V_2 היא 0 או 1 או 2. כלומר, ההבדל משידוך רגיל הוא שכל אחד מצמתי V_2 יכול להיות מחובר לשני צמתים מ- V_1 .

השאלה עוסקת בבעיית האופטימיזציה של מציאת קבוצה גדולה ביותר של קשתות המהווה semi-matching.

ספוילר להמשך הבחינה: גם שאלה 6.1 עוסקת בבעיה זו (בגרסת ה-online שלה).



1.2 (6 נק') עבור הגרף שבאיור, תארו את הבעיה בכתוב וקטורי ומטריציוני, וכן את הבעיה הדואלית המתאימה, גם היא כ-BIP. סמנו את המשתנים של הבעיה הדואלית ב- $y_a, y_b, \dots, y_e$.

— YOUR WORK —

Describe the general dual of the semi-matching BIP and identify it as weighted vertex cover.

שאלה 1 (18 נק')

יהי $G=(V,E)$ גרף דו-צדדי לא מכוון, כלומר $V=V_1\cup V_2$, $E\subseteq V_1\times V_2$. קבוצת קשתות $E'\subseteq E$ מהווה **semi-matching** אם בגרף $G=(V,E')$ דרגת כל צומת ב- V_1 היא 0 או 1, ודרגת כל צומת ב- V_2 היא 0 או 1 או 2. כלומר, ההבדל משידוך רגיל הוא שכל אחד מצמתי V_2 יכול להיות מחובר לשני צמתים מ- V_1 .

השאלה עוסקת בבעיית האופטימיזציה של מציאת קבוצה גדולה ביותר של קשתות המהווה semi-matching.

ספוילר להמשך הבחינה: גם שאלה 6.1 עוסקת בבעיה זו (בגרסת ה-online שלה).

1.3 (4 נק') תארו את הבעיה הדואלית של הבעיה הפרימאלית (הכללית, מסעיף 1.1) כבעיית BIP. הסבירו במילים מהם המשתנים, מהם האילוצים, ומהי משמעות פונק' המטרה.

— YOUR WORK —

Linear Programming — Integer Programming & TUM

31 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · Integer Programming & TUM

2023A · P2 **EASY** · 2023 Moed A exam

Write an integer program for Minimum Dominating Set and justify variables and constraints.

Let $G = (V, E)$ be an undirected graph.

A **dominating set** in G is a subset $D \subseteq V$ such that every vertex $u \in V$ is either in D or has a neighbor in D .

In the **Minimum Dominating Set** problem, the goal is to find a dominating set which is as small as possible.

Write an integer program that corresponds to this problem. Explain your answer briefly: write what are the variables and what is their meaning, and explain why every feasible solution to the IP corresponds to a solution to the original problem.

— YOUR WORK —

Write an integer linear program for weighted Max-SAT and justify variables and constraints.

In the weighted Max-SAT problem, and input is a CNF formula in which every clause C_j has a non-negative weight w_j , and the goal is to find an assignment in which the weight of the satisfied clauses is maximal. Write an integer linear program that corresponds to this problem. Explain your answer briefly: write what are the variables and what is their meaning, and explain why every feasible solution to the IP corresponds to a solution to the original problem.

— YOUR WORK —

T/F: a polynomial-time algorithm for integer linear programs would imply $P=NP$ (reduction from independent set).

Problem 1 (4×6=24 points)

For each of the following statements, write whether they are true or false, and briefly explain your answer.

(a) If there's an algorithm that solves integer linear programs in polynomial time, then $P=NP$.

— YOUR WORK —

Write an integer linear program for Minimum Makespan (assign n jobs to m machines minimizing max load).

In the Minimum Makespan problem, the input is a list of n jobs with processing times $p_1, \dots, p_n$ that need to be assigned to m machines. The machines operate in parallel, but every machine can process only one job at a time, and the goal is find an assignment in which the total time of the machine with the heaviest load is the smallest (that is, we seek to minimize the total processing time of the machine that finishes last).

Write an integer linear program that corresponds to this problem. Explain your answer briefly.

— YOUR WORK —

Write the integer linear program for weighted set cover and explain variables/constraints.

Problem 2 (10+10=20 points)

In the set cover problem, the input is a family of subsets $S_1, \dots, S_m \subseteq \{1, 2, \dots, n\}$ where every set has a weight c_i . Every $i \in \{1, 2, \dots, n\}$ appears in at least one of the sets. The goal is to find a family of sets whose union is $\{1, 2, \dots, n\}$, where the total weight is minimal (the weight of a family is the sum of weights of sets in it).

(a) Write an integer linear program that corresponds to this problem. Explain your answer briefly.

— YOUR WORK —

Prove every Hamiltonian cycle gives a feasible solution to the degree-constraint TSP integer program.

Problem 4 (6+6+6=18 points)

שאלה 4 (18=6+6+6 נקודות)

The **traveling salesman problem** asks, given a directed and weighted graph G , to find a minimum weight Hamiltonian cycle in G (a Hamiltonian cycle is a cycle that goes through each vertex exactly once. The weight of a cycle is the sum of weights of the edges in the cycle).

We'd like to express this problem as an integer program. For every edge (u, v) whose weight is $c_{u,v}$ we'll have a variable $x_{u,v}$, and we consider the program:

$$\begin{array}{ll} \text{minimize} & \sum_{(u,v) \in E} c_{u,v} x_{u,v} \\ \text{subject to} & \sum_{u: (u,v) \in E} x_{u,v} = 1 \quad \text{for all } v \in V \\ & \sum_{w: (v,w) \in E} x_{v,w} = 1 \quad \text{for all } v \in V \\ & x_{u,v} \in \{0, 1\} \quad \text{for all } (u, v) \in E \end{array}$$

בעיית הסוכן הנוסע מבקשת, בהינתן גרף מכוון וממושקל $G = (V, E)$, למצוא מעגל המילטוני בעל משקל מינימלי (מעגל המילטוני הוא מעגל המבקר בכל קדקוד בדיוק פעם אחת. משקל המעגל הוא סכום המשקלות של הקשתות במעגל).

נרצה לנסח את הבעיה כבעיית תכנון בשלמים (integer program). לכל קשת (u, v) שמשקלה $c_{u,v}$ נתאים משתנה $x_{u,v}$, ונביט בתכנית בשלמים:

(a) Prove that every Hamiltonian cycle corresponds to a feasible solution to the program above.

— YOUR WORK —

Show the converse fails: a feasible IP solution (two disjoint cycles) not corresponding to a Hamiltonian cycle.

Problem 4 (6+6+6=18 points)

שאלה 4 (18=6+6+6 נקודות)

The **traveling salesman problem** asks, given a directed and weighted graph G , to find a minimum weight Hamiltonian cycle in G (a Hamiltonian cycle is a cycle that goes through each vertex exactly once. The weight of a cycle is the sum of weights of the edges in the cycle).

We'd like to express this problem as an integer program. For every edge (u, v) whose weight is $c_{u,v}$ we'll have a variable $x_{u,v}$, and we consider the program:

$$\begin{array}{ll} \text{minimize} & \sum_{(u,v) \in E} c_{u,v} x_{u,v} \\ \text{subject to} & \sum_{u: (u,v) \in E} x_{u,v} = 1 \quad \text{for all } v \in V \\ & \sum_{w: (v,w) \in E} x_{v,w} = 1 \quad \text{for all } v \in V \\ & x_{u,v} \in \{0, 1\} \quad \text{for all } (u, v) \in E \end{array}$$

בעיית הסוכן הנוסע מבקשת, בהינתן גרף מכוון וממושקל $G = (V, E)$, למצוא מעגל המילטוני בעל משקל מינימלי (מעגל המילטוני הוא מעגל המבקר בכל קדקוד בדיוק פעם אחת. משקל המעגל הוא סכום המשקלות של הקשתות במעגל).

נרצה לנסח את הבעיה כבעיית תכנון בשלמים (integer program). לכל קשת (u, v) שמשקלה $c_{u,v}$ נתאים משתנה $x_{u,v}$, ונביט בתכנית בשלמים:

(b) Prove that the other direction **isn't** true: show a graph that has a feasible solution to the program above that doesn't correspond to a Hamiltonian cycle.
(Hint: think about connectivity)

Prove that adding subtour-elimination constraints makes every feasible IP solution a Hamiltonian cycle.

Problem 4 (6+6+6=18 points)

שאלה 4 (18=6+6+6 נקודות)

The **traveling salesman problem** asks, given a directed and weighted graph G , to find a minimum weight Hamiltonian cycle in G (a Hamiltonian cycle is a cycle that goes through each vertex exactly once. The weight of a cycle is the sum of weights of the edges in the cycle).

We'd like to express this problem as an integer program. For every edge (u, v) whose weight is $c_{u,v}$ we'll have a variable $x_{u,v}$, and we consider the program:

$$\begin{array}{ll} \text{minimize} & \sum_{(u,v) \in E} c_{u,v} x_{u,v} \\ \text{subject to} & \sum_{u: (u,v) \in E} x_{u,v} = 1 \quad \text{for all } v \in V \\ & \sum_{w: (v,w) \in E} x_{v,w} = 1 \quad \text{for all } v \in V \\ & x_{u,v} \in \{0, 1\} \quad \text{for all } (u, v) \in E \end{array}$$

בעיית הסוכן הנוסע מבקשת, בהינתן גרף מכוון וממושקל $G = (V, E)$, למצוא מעגל המילטוני בעל משקל מינימלי (מעגל המילטוני הוא מעגל המבקר בכל קדקוד בדיוק פעם אחת. משקל המעגל הוא סכום המשקלות של הקשתות במעגל).

נרצה לנסח את הבעיה כבעיית תכנון בשלמים (integer program). לכל קשת (u, v) שמשקלה $c_{u,v}$ נתאים משתנה $x_{u,v}$, ונביט בתכנית בשלמים:

(c) We'll add to the IP the following constraints

$$\sum_{u \in Q} \sum_{v \in Q: (u,v) \in E} x_{u,v} \leq |Q| - 1$$

Prove that now every feasible solution to the IP corresponds to a Hamiltonian cycle.

Write an integer linear program for bin packing (bin-used and item-assignment variables) and justify it.

Problem 4 (14 points)

In the bin packing problem, the input is a positive integer B and a set of n weighted items, such that the weight of the i -th element is w_i , and it holds that $0 \leq w_i \leq B$.

The goal is to pack the n elements in as few bins as possible, where the weight of every bin is at most B .

Write an integer linear program that corresponds to this problem. Explain your answer briefly: write what are the variables and what is their meaning, and explain why every feasible solution to the IP corresponds to a solution to the original problem.

— YOUR WORK —

Priority 2 — Homework · Integer Programming & TUM

hw2026s4 · P3 (a) **EASY** · HW 2026 · Set 4

Give the IP/LP formulation of set cover

3. In this problem we'll give an example for a *randomized rounding* scheme for the set cover problem.

In the set cover problem, we are given a collection $\mathcal{F}$ of m subsets of $\{1, 2, \dots, n\}$ such that every $1 \leq i \leq n$ is contained in some set in $\mathcal{F}$. A subfamily $\mathcal{S} \subseteq \mathcal{F}$ such that $\bigcup_{S \in \mathcal{S}} S = \{1, \dots, n\}$ is called a *cover* of $\{1, \dots, n\}$. Note that by our assumption, $\mathcal{F}$ itself is a cover. We would like to find a cover $\mathcal{S}$ such that $|\mathcal{S}|$ (i.e., the number of sets in $\mathcal{S}$) is as small as possible.

- (a) Formulate the set cover problem as an IP: use a variable x_S for each set $S \in \mathcal{F}$. One of the constraints should be that the variables take values in $\{0, 1\}$.

— YOUR WORK —

Write an integer program for maximum clique

2. Formulate each of the following NP-hard problems as an Integer Programming (IP) problem. That is, formulate an IP problem whose feasible solutions correspond to solutions of the original problem, and whose optimal solution (if applicable) corresponds to an optimal solution of the original problems. You don't need to write a formal detailed proof that the IP corresponds to the original problem. Please only describe what is the set of variables of the IP and provide a short explanation for the constraints.
- (a) MAX-CLIQUE: given an undirected graph $G = (V, E)$, find the clique of maximal size in G .

— YOUR WORK —

Write an integer program for subset-sum feasibility

2. Formulate each of the following NP-hard problems as an Integer Programming (IP) problem. That is, formulate an IP problem whose feasible solutions correspond to solutions of the original problem, and whose optimal solution (if applicable) corresponds to an optimal solution of the original problems. You don't need to write a formal detailed proof that the IP corresponds to the original problem. Please only describe what is the set of variables of the IP and provide a short explanation for the constraints.
- (b) SUBSET-SUM: given a set of integers $U = \{w_1, \dots, w_n\}$ and a target t , decide if there's a subset $S \subseteq U$ such that $\sum_{w \in S} w = t$.

— YOUR WORK —

Write an integer program for graph coloring / chromatic number

2. Formulate each of the following NP-hard problems as an Integer Programming (IP) problem. That is, formulate an IP problem whose feasible solutions correspond to solutions of the original problem, and whose optimal solution (if applicable) corresponds to an optimal solution of the original problems. You don't need to write a formal detailed proof that the IP corresponds to the original problem. Please only describe what is the set of variables of the IP and provide a short explanation for the constraints.
- (c) MIN-COLORING: given an undirected graph $G = (V, E)$, find a *legal coloring* of the vertices of G that uses as few colors as possible. A coloring of the vertices of G is a function $c : V \rightarrow \{1, \dots, n\}$ that assigns a color for each vertex. A coloring is *legal* if for every edge $(u, v) \in E$, $c(u) \neq c(v)$.

— YOUR WORK —

Identify the graph problem of a given covering ILP (vertex cover).

8. Let V be a finite set. Let E be a set of unordered pairs of elements in V .

a. Consider the following integer program IP1:

$$\min \sum_{v \in V} x_v \text{ s.t.}$$

$$x_u + x_v \geq 1 \text{ for all } uv \in E$$

$$x_v \in \{0,1\} \text{ for all } v \in V$$

What is the name of the corresponding graph theoretic problem?

— YOUR WORK —

Write the LP relaxation of the vertex-cover ILP.

8. Let V be a finite set. Let E be a set of unordered pairs of elements in V .
- b. Write down LP1, the relaxation of IP1.

— YOUR WORK —

Write minimum hitting set as a binary integer linear program.

- a. (5 points) Let S be a finite set. Let C be a collection of subsets of S . A hitting set of C is a subset $X \subseteq S$ that contains at least one element from every subset D in C ($\forall D \in C \quad X \cap D \neq \emptyset$). The minimum hitting set problem is the problem of finding a hitting set X of minimum cardinality. Write down the minimum hitting set problem as a binary integer linear program.

— YOUR WORK —

Write the knapsack problem as a binary integer linear program.

b) (5 points) Write down the knapsack problem as a binary linear program (BIP).

— YOUR WORK —

Write the LP relaxation of the knapsack BIP.

c) (3 points) What is the relaxation of the BIP from (b)?

— YOUR WORK —

Write minimum-cost perfect matching as an integer linear program, defining the variables.

3. (17 points) Let $G=(V,E)$ be an undirected graph with a non-negative edge weight $w(e)$ for each edge e in E . A matching in G is a subset M of the edges E such that no two edges in M are incident to the same vertex. A perfect matching is a matching M in which every vertex $v \in V$ is incident to some edge in M .
- a) (5 points) Write down the problem of finding a minimum cost perfect matching as an integer linear program. Clearly state what are the variables and what is their meaning

— YOUR WORK —

Write the LP relaxation of the perfect-matching ILP, removing redundant constraints.

- b) (2 points) Write down the LP relaxation of the integer program from item a. Remove any redundant constraints.

— YOUR WORK —

Write a 0/1 BIP to select maximum students subject to at most c_j selected per course; explain objective and constraints.

שאלה 2 (18 נק')

נתונה קבוצה של n סטודנטים $A = \{a_1, a_2, \dots, a_n\}$ ו- m קורסים. לכל $1 \leq j \leq m$, בקורס ה- j משתתפת קבוצת הסטודנטים $U_j \subseteq A$. בנוסף נתון לכל קורס פרמטר c_j . עליכם לבחור קבוצה גדולה ביותר של סטודנטים תחת המגבלה שלכל קורס $1 \leq j \leq m$ יהיו בקבוצת הסטודנטים הנבחרת לכל היותר c_j נציגים מתוך U_j . יתכן שסטודנט שנבחר משתתף במספר קורסים, ואז הוא נספר כנציג עבור כל אחד מהם.

2.1 (4 נק') לכל $1 \leq i \leq n$, יהי x_i משתנה בינארי המתאר האם הסטודנט ה- i נבחר. תארו את הבעיה כבעיית תכנות לינארי 0/1 (BIP). הסבירו במילים את פונק' המטרה והאילוצים.

— YOUR WORK —

Formulate maximum semi-matching in a bipartite graph as a 0/1 binary integer program; explain objective/constraints.

שאלה 1 (18 נק')

יהי $G=(V,E)$ גרף דו-צדדי לא מכוון, כלומר $V=V_1\cup V_2$, $E\subseteq V_1\times V_2$. קבוצת קשתות $E'\subseteq E$ מהווה **semi-matching** אם בגרף $G=(V,E')$ דרגת כל צומת ב- V_1 היא 0 או 1, ודרגת כל צומת ב- V_2 היא 0 או 2. כלומר, ההבדל משידוך רגיל הוא שכל אחד מצמתי V_2 יכול להיות מחובר לשני צמתים מ- V_1 .

השאלה עוסקת בבעיית האופטימיזציה של מציאת קבוצה גדולה ביותר של קשתות המהווה semi-matching.

ספוילר להמשך הבחינה: גם שאלה 6.1 עוסקת בבעיה זו (בגרסת ה-online שלה).

1.1 (4 נק') לכל קשת $e\in E$, יהי x_e משתנה בינארי המתאר האם הקשת e נבחרה לפתרון. תארו את הבעיה כבעיית תכנות לינארי (BIP) 0/1. הסבירו במילים את פונק' המטרה והאילוצים.

— YOUR WORK —

Write minimum weight dominating set as an integer linear program.

4. (8 points) Recall the weighted dominating set problem. The input is a graph $G=(V,E)$ with vertex weights w_v . A set S of vertices dominates a vertex v if $v \in S$ or if v has a neighbor in S . The output is a minimum weight subset of V that dominates all the vertices in V .
- a. (5 points) Write down the weighted dominating set as an integer linear program.

— YOUR WORK —

Give a dominating-set instance where the LP relaxation optimum is fractional (integrality gap).

4. (8 points) Recall the weighted dominating set problem. The input is a graph $G=(V,E)$ with vertex weights w_v . A set S of vertices dominates a vertex v if $v \in S$ or if v has a neighbor in S . The output is a minimum weight subset of V that dominates all the vertices in V .
- (5 points) Write down the weighted dominating set as an integer linear
 - (3 points) Give an instance where an optimal solution to the LP relaxation of the ILP in (a.) is not integral.

— YOUR WORK —

Write bin packing as an integer linear program and explain the variables.

7. Recall the bin packing problem. The input is a list of items with weights w_i . The goal is to pack the items into the smallest number of bins, such that the total weight packed in each bin is at most 1.
 - a. Write the problem as an integer linear program. Shortly explain the meaning of the variables you choose.

— YOUR WORK —

Identify the graph problem of a given ILP (maximum independent set).

7. (23 points) Let $G=(V,E)$ be an undirected connected graph with $|V|=n$.

a. (3 points) What is the graph problem described by the following ILP?

$$\begin{aligned} \max \quad & \sum_{v \in V} x_v \quad \text{s.t.} \\ x_v + x_u & \leq 1 \quad \forall (u, v) \in E \\ x_v & \in \{0,1\} \quad \forall v \in V \end{aligned}$$

— YOUR WORK —

Write the LP relaxation of the maximum-independent-set ILP.

b. (2 points) Write down the LP relaxation of the above ILP.

— YOUR WORK —

Prove the independent-set LP optimum is at least $n/2$ via the all-halves feasible solution.

c. (3 points) Prove that the optimal solution of the LP in (b) is at least $n/2$.

The solution $x_v = \frac{1}{2}$ for all $v \in V$ is a feasible solution whose objective value is $n/2$

— YOUR WORK —

Describe a 100-vertex graph whose independent-set ILP optimum is 1 (a clique).

- d. (3 points) Describe a graph with 100 vertices such that the optimal solution to the ILP in (a) is 1.

— YOUR WORK —

Poly-time algorithm for minimum edge cover via maximum matching, with a correctness argument.

- g. (5 points) Describe a polynomial time algorithm that solves the ILP in (f). Shortly explain why your algorithm works. You do not have to specify the exact running time. You may use in your answer any algorithm mentioned in class.

— YOUR WORK —

Write maximum-weight matching as a binary integer linear program.

6. (25 points)
- a. (5 points) Let $G=(V,E)$ be an undirected graph with edge weight $w(e)$ for each edge $e \in E$. Write down the problem of finding a maximum weight matching in G as a binary integer linear program.

— YOUR WORK —

Approximation — Approximation Algorithms

34 questions · solutions & a study guide for this topic are on the website

Priority 1 — 2023+ Exams · Approximation Algorithms

2023A · P3 **MEDIUM** · 2023 Moed A exam

Give a $2/3$ -approximation for Maximum Triple Cut (random 3-partition, derandomized via conditional expectations).

In the **Maximum Triple Cut** problem, the input is an undirected graph $G = (V, E)$, and we would like to find a partition of the vertex set V into three disjoint sets V_1, V_2, V_3 such that the number of edges whose two endpoints are in different sets is as large as possible.

Give an efficient algorithm that obtains a $2/3$ -approximation for the Maximum Triple Cut problem.

— YOUR WORK —

Give a 3-approximation for vertex cover in a 3-uniform hypergraph (greedy or LP rounding).

Problem 3 (16 points)

A **3-uniform hypergraph** $G = (V, E)$ consists of a set of vertices V and a set of "edges" E , such that each edge contains exactly 3 vertices (as opposed to two in a "normal" graph).

In the **vertex cover** problem, the input is an 3-uniform hypergraph $G = (V, E)$, and we would like to find set of minimum size such that every edge in the graph contains a vertex from C .

Give an efficient algorithm that obtains a 3-approximation for the vertex cover problem in a 3-uniform hypergraph.

— YOUR WORK —

d -approximation for minimum-weight representative set (hitting set with $|T_j| \leq d$) via LP relaxation and rounding.

Problem 5 (16 points)

שאלה 5 (16 נקודות)

Let $V = \{1, 2, \dots, n\}$ be a set of n elements, in which the i -th element has a weight w_i . Let $T_1, \dots, T_k$ be subsets of V where for every j , $|T_j| \leq d$.

A subset $H \subseteq V$ is called **representative** if it contains, for every $1 \leq j \leq k$, at least one element from T_j .

Give a d -approximation algorithm for the problem of finding a minimum weight representative set, based on relaxation and rounding of the following integer linear program:

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^n w_i x_i \\ \text{subject to} & \sum_{i \in T_j} x_i \geq 1 \quad \text{for all } 1 \leq j \leq k \\ & x_i \in \{0, 1\} \quad \text{for all } 1 \leq i \leq n \end{array}$$

תהי $V = \{1, 2, \dots, n\}$ קבוצה בת n איברים, כאשר לאיבר ה- i יש משקל w_i . תהיינה $T_1, \dots, T_k$ קבוצות של V , כאשר לכל j , $|T_j| \leq d$. תת קבוצה $H \subseteq V$ תקרא **מייצגת** אם היא מכילה, לכל $1 \leq j \leq k$, לפחות איבר אחד מ- T_j . תנו אלגוריתם d -קירוב לבעיית מציאת הקבוצה המייצגת בעלת משקל מינימלי, המבוסס על רלקסציה ועיגול של התכנית הלינארית בשלמים הבאה:

— YOUR WORK —

Give an f -approximation for set cover of frequency f via LP relaxation and threshold rounding at $1/f$.

Problem 2 (10+10=20 points)

In the set cover problem, the input is a family of subsets $S_1, \dots, S_m \subseteq \{1, 2, \dots, n\}$ where every set has a weight c_i . Every $i \in \{1, 2, \dots, n\}$ appears in at least one of the sets. The goal is to find a family of sets whose union is $\{1, 2, \dots, n\}$, where the total weight is minimal (the weight of a family is the sum of weights of sets in it).

(b) In the set cover problem with frequency f , every $i \in \{1, 2, \dots, n\}$ is guaranteed to appear in at most f sets.

Give an f -approximation algorithm to the set cover problem with frequency f , based on relaxation and rounding of the integer linear program from item (a).

— YOUR WORK —

Prove the sets whose dual constraints are tight form a set cover (complementary-slackness / primal-dual).

Problem 2 (10+10=20 points)

שאלה 2 (20=10+10 נקודות)

In the set cover problem, the input is a family of subsets $S_1, \dots, S_m \subseteq \{1, 2, \dots, n\}$ where every set has a weight c_i . Every $j \in \{1, 2, \dots, n\}$ appears in at least one of the sets. The goal is to find a family of sets whose union is $\{1, 2, \dots, n\}$, where the total weight is minimal (the weight of a family is the sum of weights of sets in it).

Recall the integer LP that corresponds to this problem:

בבעיית הכיסוי בקבוצות הקלט הוא משפחה של תתי קבוצות $S_1, \dots, S_m \subseteq \{1, 2, \dots, n\}$ כאשר לכל קבוצה משקל c_i . כל איבר $j \in \{1, 2, \dots, n\}$ מופיע בלפחות אחת מהקבוצות. המטרה היא למצוא משפחה של קבוצות שאיחודה הוא $\{1, 2, \dots, n\}$ עם משקל כולל מינימלי (משקל של משפחה הוא סכום משקלי הקבוצות שבה).

ניזכר בתכנית הלינארית בשלמים המתאימה לבעיה:

$$\begin{aligned} & \text{minimize} && \sum_{i=1}^m c_i x_i \\ & \text{subject to} && \sum_{i: j \in S_i} x_i \geq 1 \quad \text{for every } 1 \leq j \leq n \\ & && x_i \in \{0, 1\} \quad \text{for every } 1 \leq i \leq m \quad (1) \end{aligned}$$

(b) Let y^* be an **optimal** solution for (D). Consider the family of subsets S_i such that the constraint (2) holds with equality, that is, $\sum_{j: j \in S_i} y_j^* = c_i$. Show that this family of sets is a set cover.

— YOUR WORK —

Show an f -approximation for set cover by solving the dual and picking tight sets (primal-dual analysis).

Problem 2 (10+10=20 points)

שאלה 2 (20=10+10 נקודות)

In the set cover problem, the input is a family of subsets $S_1, \dots, S_m \subseteq \{1, 2, \dots, n\}$ where every set has a weight c_i . Every $j \in \{1, 2, \dots, n\}$ appears in at least one of the sets. The goal is to find a family of sets whose union is $\{1, 2, \dots, n\}$, where the total weight is minimal (the weight of a family is the sum of weights of sets in it).

Recall the integer LP that corresponds to this problem:

בבעיית הכיסוי בקבוצות הקלט הוא משפחה של תתי קבוצות $S_1, \dots, S_m \subseteq \{1, 2, \dots, n\}$ כאשר לכל קבוצה משקל c_i . כל איבר $j \in \{1, 2, \dots, n\}$ מופיע בלפחות אחת מהקבוצות. המטרה היא למצוא משפחה של קבוצות שאיחודה הוא $\{1, 2, \dots, n\}$ עם משקל כולל מינימלי (משקל של משפחה הוא סכום משקלי הקבוצות שבה).

ניזכר בתכנית הלינארית בשלמים המתאימה לבעיה:

$$\begin{array}{ll} \text{minimize} & \sum_{i=1}^m c_i x_i \\ \text{subject to} & \sum_{i: S_i \ni j} x_i \geq 1 \quad \text{for every } 1 \leq j \leq n \\ & x_i \in \{0, 1\} \quad \text{for every } 1 \leq i \leq m \end{array} \quad (1)$$

(c) In the set cover problem with frequency f , every $i \in \{1, 2, \dots, n\}$ is guaranteed to appear in at most f sets.

Show an efficient f -approximation algorithm to the set cover problem with frequency f , based on solving (D) and picking a set cover as described in item (b).

Priority 2 — Homework · Approximation Algorithms

hw2025s5 · P2 **MEDIUM** · HW 2025 · Set 5

Greedy $1/(\Delta+1)$ -approximation for maximum independent set on graphs of max degree Δ

Corollary 0.3. *The algorithm above is a $\frac{1}{\Delta+1}$ -approximation algorithm for the Max-Independent-Set problem on graphs where the maximal degree is at most Δ .*

— YOUR WORK —

Randomized LP rounding for set cover: show $E[|S|] = LPOPT \leq OPT$

3. In this problem we'll give an example for a *randomized rounding* scheme for the set cover problem.

In the set cover problem, we are given a collection $\mathcal{F}$ of m subsets of $\{1, 2, \dots, n\}$ such that every $1 \leq i \leq n$ is contained in some set in $\mathcal{F}$. A subfamily $\mathcal{S} \subseteq \mathcal{F}$ such that $\bigcup_{S \in \mathcal{S}} S = \{1, \dots, n\}$ is called a *cover* of $\{1, \dots, n\}$. Note that by our assumption, $\mathcal{F}$ itself is a cover. We would like to find a cover $\mathcal{S}$ such that $|\mathcal{S}|$ (i.e., the number of sets in $\mathcal{S}$) is as small as possible.

- (b) Show that the *expected* size of $\mathcal{S}$ equals $LPOPT$ (which is at most OPT , the minimum size of a set cover).

— YOUR WORK —

Show each element is covered with probability $\geq 1-1/e$ under one round of LP rounding

3. In this problem we'll give an example for a *randomized rounding* scheme for the set cover problem.

In the set cover problem, we are given a collection $\mathcal{F}$ of m subsets of $\{1, 2, \dots, n\}$ such that every $1 \leq i \leq n$ is contained in some set in $\mathcal{F}$. A subfamily $\mathcal{S} \subseteq \mathcal{F}$ such that $\bigcup_{S \in \mathcal{S}} S = \{1, \dots, n\}$ is called a *cover* of $\{1, \dots, n\}$. Note that by our assumption, $\mathcal{F}$ itself is a cover. We would like to find a cover $\mathcal{S}$ such that $|\mathcal{S}|$ (i.e., the number of sets in $\mathcal{S}$) is as small as possible.

- (c) We say i is *covered* by $\mathcal{S}$ if there's $S \in \mathcal{S}$ such that $i \in S$. Prove that every $1 \leq i \leq n$ is covered by $\mathcal{S}$ with probability at least $1 - 1/e$.

— YOUR WORK —

Repeat rounding $T = \ln(n/\epsilon)$ times; union bound gives an $O(\log n)$ set-cover approximation

3. In this problem we'll give an example for a *randomized rounding* scheme for the set cover problem.

In the set cover problem, we are given a collection $\mathcal{F}$ of m subsets of $\{1, 2, \dots, n\}$ such that every $1 \leq i \leq n$ is contained in some set in $\mathcal{F}$. A subfamily $\mathcal{S} \subseteq \mathcal{F}$ such that $\bigcup_{S \in \mathcal{S}} S = \{1, \dots, n\}$ is called a *cover* of $\{1, \dots, n\}$. Note that by our assumption, $\mathcal{F}$ itself is a cover. We would like to find a cover $\mathcal{S}$ such that $|\mathcal{S}|$ (i.e., the number of sets in $\mathcal{S}$) is as small as possible.

- (d) Show that for $T = \ln(n/\epsilon)$, $\mathcal{C}$ is a cover with probability at least $1 - \epsilon$ and the size of $\mathcal{C}$ in expectation is at most $\text{OPT} \cdot \ln(n/\epsilon)$.

— YOUR WORK —

SDP relaxation + Goemans-Williamson random-hyperplane rounding gives a 0.878 approx for Max-2-LIN

4. In this question we'll consider the problem Max-2-LIN which is defined as follows: we have a set of n variables $x_1, \dots, x_n$, each taking values in $\{-1, 1\}$. Each constraint is either of the form $x_i x_j = 1$ (which is equivalent to $x_i = x_j$) or $x_i x_j = -1$ (which is equivalent to $x_i \neq x_j$). We'd like to find assignment to $x_1, \dots, x_n$ which maximizes the number of satisfied constraints.

Note that if we only had constraints of the form $x_i x_j = -1$ then this is identical to the Max-Cut problem we talked about in class, but now we're considering a more general problem. Show that this problem also has a $0.878\dots$ approximation algorithm (hint: use the same SDP method we saw in class, just tweak the analysis a bit).

— YOUR WORK —

Priority 3 — Pre-2023 Exams · Approximation Algorithms

2017A · P1 (e) **EASY** · 2017 Moed A exam

True/False: an algorithm with $(1+\epsilon)$ -quality and runtime $O(n^{3+1/\epsilon})$ is a PTAS.

- e. True or False: “Let Q be a minimization problem. Suppose that, for any number $0 < \epsilon < 1$ there is an algorithm A that, given an instance x of Q of size n , produces a solution whose quality is at most $(1 + \epsilon)$ times the quality of an optimal solution to x . The running time of A is $O(n^{3+1/\epsilon})$. A is a polynomial time approximation scheme (PTAS) for problem Q .”
Give a short explanation.

— YOUR WORK —

2-approx for min-cost subgraph connecting each receiver to some sender, via reduction to Steiner tree.

5. (8 points) Let $G=(V,E)$ be an undirected graph with non-negative edge costs. Let S (senders) and R (receivers) be two disjoint subsets of V , such that $S \cup R \neq V$. Give a 2-approximation algorithm for the problem of finding a minimum cost subgraph of G in which every receiver has a path to some sender (not necessarily the same sender for all receivers). You may use any algorithm seen in class or in the problem sets.

— YOUR WORK —

MAX-3SAT: random assignment gives 7/8-approximation in expectation, prove via linearity of expectation.

8. (14 points) Recall the MAX-3SAT problem. The input is a 3CNF formula with m clauses over n variables. The output is an assignment of truth values to the variables that maximizes the number of satisfied clauses. You may assume that the no 2 literals in the same clause belong to the same variable.
- a. (7 points) Give a randomized algorithm that is a 7/8-approximation algorithm in expectation. Prove your answer.

— YOUR WORK —

Prove any maximal matching is a 2-approximation for maximum matching.

9. (21 points) Recall that a *matching* in a graph $G=(V,E)$ is a set of edges that share no vertices. A matching M is maximal if there is no matching N such that $M \subset N$. A matching M is a maximum matching if for every matching N , $|M| \geq |N|$.
- a. (7 points) Prove that a maximal matching is a 2-approximation for the maximum matching.

— YOUR WORK —

2-approximation for minimizing bin capacity with fixed number of bins (greedy least-loaded scheduling).

7. Recall the bin packing problem. The input is a list of items with weights w_i . The goal is to pack the items into the smallest number of bins, such that the total weight packed in each bin is at most 1.
- b. Consider the variant in which the number of available bins is given as input and the goal is to find the minimum bin capacity (all bins have the same capacity) such that all the items can be packed into the available bins. Give a 2-approximation algorithm for this variant.

— YOUR WORK —

Prove min-weight maximum matching $MATCH \leq OPT/2$ for the TSP tour in a complete weighted graph.

3. (10 points) Let G be a complete graph with edge weights. Let OPT be the weight of a minimum traveling salesperson tour in G . Let $MATCH$ be the weight of a minimum weight maximum matching in G . Prove $MATCH \leq OPT/2$.

— YOUR WORK —

Prove or refute that a given greedy algorithm is a 2-approximation for maximum cut (refute).

5. (10 points) Recall that in the maximum cut problem, given an undirected graph $G=(V,E)$ we want to partition V into two nonempty disjoint sets $V = S \cup T$, such that the number of edges that have one endpoint in S and one endpoint in T is maximized. Consider the following greedy algorithm:

```
Let uv be an arbitrary edge
set  $S = \{\}$ ,  $T = \{\}$ 
For each vertex  $w$  in  $V$  (in arbitrary order):
    if  $w$  has more neighbors at  $S$  than at  $T$  add  $w$  to  $S$ 
    else add  $w$  to  $T$ 
End for
Return  $(S,T)$ 
```

Prove or refute: "The above algorithm is a 2-approximation for maximum cut".

— YOUR WORK —

Prove the greedy ratio-sorted knapsack algorithm has unbounded approximation ratio (counterexample).

2. (20 points)

- a) (7 points) Recall the knapsack problem. The input is a set of items, and a number W . Item i has a cost b_i and a weight w_i . The goal is to find a set S of items with maximum total cost, whose total weight is at most W . We discussed in class an approximation algorithm that sorts the items according to the ratio b_i/w_i , and greedily adds items to the set S in descending order. Prove that the approximation ratio of this algorithm is not bounded.

— YOUR WORK —

Describe a polynomial 3-approximation for minimum hitting set when every subset has size ≤ 3 .

2. (20 points) Let S be a finite set. Let C be a collection of subsets of S . A hitting set of C is a subset $X \subseteq S$ that contains at least one element from every subset D in C ($\forall D \in C \quad X \cap D \neq \emptyset$). The minimum hitting set problem is the problem of finding a hitting set X of minimum cardinality.
- a) (5 points) Suppose that each subset D in C has cardinality at most 3 ($\forall D \in C \quad |D| \leq 3$). Describe a polynomial 3-approximation algorithm for minimum hitting set in this case. Only describe your algorithm (either in pseudocode or in free text). Make sure to define any non-standard notation you are using. Your description should be concise, accurate and complete.

— YOUR WORK —

Prove the $|D| \leq 3$ hitting-set greedy is a polynomial 3-approximation (disjoint-sets lower bound).

- b) (10 points) Prove that your algorithm is a polynomial time 3-approximation algorithm for minimum hitting set. Your proof should be concise, accurate, and complete.

— YOUR WORK —

Give a 3-competitive online algorithm for hitting set with $|D| \leq 3$ (same greedy works in arrival order).

- c) (5 points) In the online version of the problem the subsets are revealed one by one, and the online algorithm maintains a hitting set X . When a subset D arrives, the online algorithm must guarantee that X is a hitting set for the subsets seen so far. The algorithm may add elements of S to X to guarantee this requirement. Once an element is added to X it cannot be removed. The goal is to end up with the smallest possible X . Give a 3-competitive algorithm for the case where each subset D has cardinality at most 3. You do not need to prove correctness or analyze the running time.

— YOUR WORK —

Refute that min-degree greedy independent set has approximation ratio 1 (counterexample graph).

2. (15 points) Recall that an independent set in an undirected graph $G=(V,E)$ is a subset S of V such that for every $u, v \in S$, the edge uv is not in E . Consider the following greedy algorithm for finding a maximum cardinality independent set:
1. $S = \text{empty set}$
 2. while G is not empty:
 3. let v be a vertex with minimum degree in G
 4. $S = S \cup \{v\}$
 5. remove v and its neighbors from G
 6. return S
- a. (5 points) Prove or refute: this is a polynomial time algorithm with approximation ratio 1. Make sure your answer is complete, accurate and concise.

— YOUR WORK —

Prove the min-degree greedy is a polynomial $(D+1)$ -approximation for independent set on max-degree- D graphs.

b. (10 points) Prove or refute: this algorithm is a polynomial time $\Delta+1$ approximation algorithm when the degree of every vertex of G is at most Δ . Make sure your answer is complete, accurate and concise.

— YOUR WORK —

Prove/disprove: LP-relaxation rounding ($x_i \geq 1/u \rightarrow 1$) of the student-selection LP always yields a feasible integral solution (false).

שאלה 2 (18 נק')

נתונה קבוצה של n סטודנטים $A = \{a_1, a_2, \dots, a_n\}$ ו- m קורסים. לכל $1 \leq j \leq m$, בקורס ה- j משתתפת קבוצת הסטודנטים $U_j \subseteq A$. בנוסף נתון לכל קורס פרמטר c_j . עליכם לבחור קבוצה גדולה ביותר של סטודנטים תחת המגבלה שלכל קורס $1 \leq j \leq m$ יהיו בקבוצת הסטודנטים הנבחרת לכל היותר c_j נציגים מתוך U_j . יתכן שסטודנט שנבחר משתתף במספר קורסים, ואז הוא נספר כנציג עבור כל אחד מהם.

2.4 (4 נק') נניח שלכל הקורסים $c_j = 1$ ובנוסף, בכל הקורסים יש אותו מספר סטודנטים, כלומר קיים קבוע u כך שלכל $1 \leq j \leq m$ מתקיים ש- $|U_j| = u$. פותרים את ה-LP-relaxation של הבעיה הפרימאלית מסעיף 2.1. בהנתן פתרון שבור אופטימלי, מעגלים את המשתנה x_i המתאים לסטודנט a_i באופן הבא: אם $x_i \geq \frac{1}{u}$, אז x_i מעוגל ל-1, ואחרת, x_i מעוגל ל-0.

הוכיחו או הפריכו: לכל קלט, הפתרון השלם המתקבל מאלגוריתם העיגול שתואר לעיל הוא פיזיבילי.

— YOUR WORK —

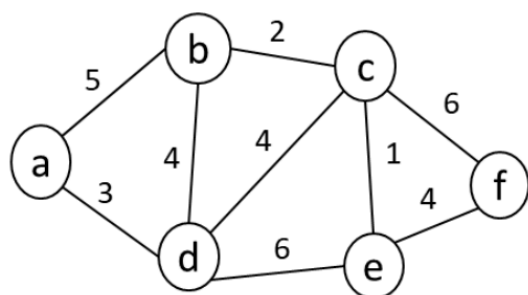
Design a $4/3$ -approximation for TSP on a complete graph with edge weights 1 or 2, using a minimum cycle cover (first step given).

שאלה 3 (16 נק').

הגדרה: **כיסוי במעגלים** של גרף לא מכוון $G=(V,E)$, הוא אוסף מעגלים פשוטים המורכבים מקשתות ב- E , כך שכל צומת ב- V משתתף במעגל אחד בדיוק.

הבהרה: מעגל פשוט הוא סדרת צמתים $v_1-v_2-\dots-v_k-v_1$ כך ש- $k \geq 3$ ולכל $j \neq 1$, הצומת v_j מופיע בסדרה רק פעם אחת. הבחירה בצומת הראשון (שהוא גם האחרון) היא שרירותית.

ידוע שקיים אלגוריתם פולינומי לפתרון הבעיה הבאה:
קלט: גרף לא מכוון $G=(V,E)$ עם משקלים על הקשתות.
פלט: האם קיים ל- G כיסוי במעגלים, ואם כן, כיסוי שמשקלו הכולל מינימלי.



דוגמה: לגרף שבאיור יש כיסוי במעגלים שמכיל את שני המעגלים $(a-d-b-a)$, $(c-e-f-c)$ ומשקלו $12+11=23$, וגם כיסוי במעגלים שמכיל את המעגל $(a-d-e-f-c-b-a)$ ומשקלו 26. האלגוריתם יחזיר אם הכיסוי הראשון.

יהי $G=(V,E)$ גרף מלא של כל הקשתות בו משקל 1 או 2.

3.1 (8 נק') הציעו אלגוריתם $\frac{4}{3}$ -קרוב לבעיית הסוכן הנוסע (TSP) עבור G .

הצעד הראשון באלגוריתם הקירוב נתון, השלימו את ההמשך

א. הרץ את האלגוריתם לעיל על G . מכיוון ש- G גרף מלא יש לו בודאות כיסוי במעגלים.
יהי C אוסף המעגלים המהווים כיסוי במעגלים במחיר מינימלי ב- G .

— YOUR WORK —

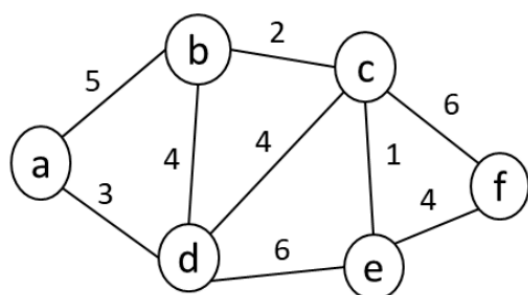
Prove the cycle-cover-based algorithm returns a TSP tour of cost at most $(4/3)OPT$.

שאלה 3 (16 נק').

הגדרה: **כיסוי במעגלים** של גרף לא מכוון $G=(V,E)$, הוא אוסף מעגלים פשוטים המורכבים מקשתות ב- E , כך שכל צומת ב- V משתתף במעגל אחד בדיוק.

הבהרה: מעגל פשוט הוא סדרת צמתים $v_1-v_2-\dots-v_k-v_1$ כך ש- $k \geq 3$ ולכל $j \neq 1$, הצומת v_j מופיע בסדרה רק פעם אחת. הבחירה בצומת הראשון (שהוא גם האחרון) היא שרירותית.

ידוע שקיים אלגוריתם פולינומי לפתרון הבעיה הבאה:
קלט: גרף לא מכוון $G=(V,E)$ עם משקלים על הקשתות.
פלט: האם קיים ל- G כיסוי במעגלים, ואם כן, כיסוי שמשקלו הכולל מינימלי.



דוגמה: לגרף שבאיור יש כיסוי במעגלים שמכיל את שני המעגלים $(a-d-b-a)$, $(c-e-f-c)$ ומשקלו $12+11=23$, וגם כיסוי במעגלים שמכיל את המעגל $(a-d-e-f-c-b-a)$ ומשקלו 26. האלגוריתם יחזיר אם הכיסוי הראשון.

3.2. (8 נק') הוכיחו שהאלגוריתם מחזיר פתרון שמשקלו לכל היותר $\frac{4}{3} \cdot OPT$ לבעיית הסוכן הנוסע.

— YOUR WORK —

Prove/disprove that rounding the LP relaxation ($x_e \geq 1/2 \rightarrow 1$) always yields a feasible integer solution (false).

שאלה 1 (18 נק')

יהי $G=(V,E)$ גרף דו-צדדי לא מכוון, כלומר $V=V_1 \cup V_2, E \subseteq V_1 \times V_2$. קבוצת קשתות $E' \subseteq E$ מהווה **semi-matching** אם בגרף $G=(V,E')$ דרגת כל צומת ב- V_1 היא 0 או 1, ודרגת כל צומת ב- V_2 היא 0 או 1, כלומר, ההבדל משידוך רגיל הוא שכל אחד מצמתי V_2 יכול להיות מחובר לשני צמתים מ- V_1 .

השאלה עוסקת בבעיית האופטימיזציה של מציאת קבוצה גדולה ביותר של קשתות המהווה semi-matching.

ספוילר להמשך הבחינה: גם שאלה 6.1 עוסקת בבעיה זו (בגרסת ה-online שלה).

1.4 (4 נק') פותרים את ה-LP-relaxation של הבעיה הפרימאלית מסעיף 1.1. בהנתן פתרון שבור אופטימלי, מעגלים את המשתנה x_e המתאים לקשת e באופן הבא: אם $x_e \geq \frac{1}{2}$, אז x_e מעוגל ל-1, ואחרת, x_e מעוגל ל-0.

הוכיחו או הפריכו: לכל קלט, הפתרון השלם המתקבל מאלגוריתם העיגול שתואר לעיל הוא פיזיבילי.

— YOUR WORK —

Give a $2/3$ -approximation for MAX-3-COLORING via sequential greedy vertex placement minimizing illegal edges.

שאלה 3 (16 נק')

נתון גרף לא מכוון $G = (V, E)$ עם פונק' משקל חיובית $w: E \rightarrow \mathbb{R}^+$ על הקשתות. 3-צביעה של צמתי הגרף היא השמה $c: V \rightarrow \{a, b, c\}$ (כל צומת מקבל את אחד מהצבעים a, b, c). בהינתן צביעה, קשת $(u, v) \in E$ צבועה באופן חוקי אם מתקיים ש- $c(u) \neq c(v)$.

בבעיית **max-3-col** המטרה היא למצוא 3-צביעה שבה משקלן הכולל של הקשתות הצבועות באופן חוקי הוא מקסימלי.

3.1 (8 נק') הציעו אלגוריתם המוצא $\frac{3}{2}$ -קירוב עבור הבעיה (כלומר, על האלג' להחזיר צביעה שבה משקלן הכולל של הקשתות הצבועות באופן חוקי הוא לפחות $\frac{2}{3}$ ממשקלן הכולל של הקשתות הצבועות באופן חוקי בצביעה האופטימלית).

הדרכה: שימו לב שצביעה בשלושה צבעים מגדירה חלוקה של צמתי הגרף ל-3 קבוצות זרות, A, B, C המכילות את הצמתיים הצבועים a, b, c בהתאמה. קשת צבועה באופן חוקי אם קצותיה שייכים לקבוצות שונות. האלגוריתם יצבע את צמתי הגרף בזה אחר זה.

— YOUR WORK —

Prove the $2/3$ approximation ratio for the greedy MAX-3-COLORING algorithm.

שאלה 3 (16 נק')

נתון גרף לא מכוון $G = (V, E)$ עם פונק' משקל חיובית $w: E \rightarrow \mathbb{R}^+$ על הקשתות.
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בבעיית **max-3-col** המטרה היא למצוא 3-צביעה שבה משקלן הכולל של הקשתות הצבועות באופן חוקי הוא מקסימלי.

3.2. (8 נק') הוכיחו את יחס הקירוב של האלגוריתם.

— YOUR WORK —

Bin packing with $1/m$ weights: prove at most $k-1$ non-full bins after packing all items of weight $\geq 1/k$.

6. (16 points) Consider a variant of the bin packing problem in which the weights of each item is of the form $1/m$, where m is integer. For example, the input may be $1/50, 1/3, 1/20, 1/7, 1/2$. Consider an algorithm that handles the items in order of decreasing weight. The algorithm places an item in any bin that can accommodate it, or in a new bin if no suitable bin exists.
- a. (7 points) Prove that after packing all items of weight at least $1/k$, there are at most $k-1$ bins that are not full.

— YOUR WORK —

Bin packing with $1/m$ weights: lower bound $(1-1/k)$ on the total weight in each non-last bin.

- b. (2 points) Suppose that, in an execution of the algorithm, the last element that required opening a new bin weighed $1/k$. Give a lower bound on the weight of items packed in every bin other than the one that was opened last.

— YOUR WORK —

Prove the decreasing-weight bin-packing algorithm is an additive 1-approximation.

- c. (7 points) Prove that this algorithm is an *additive* 1-approximation to the problem (i.e., that the algorithm uses at most one bin more than the minimum number required). Hint: use items (a) and (b) in your argument.

— YOUR WORK —

Prove a Christofides variant (match all-but-two odd vertices) is a 1.5-approx for metric Path-TSP.

3. (10 points) In the metric Path-TSP problem you are given a complete undirected graph G with non-negative edge weights satisfying the triangle inequality, and the goal is to find a minimum weight simple path that visits all nodes (i.e., a minimum cost Hamilton path). Consider the following tweak on Christofides algorithm. Let T be a minimum spanning tree of G . Let S be the set of vertices with odd degree in T . Instead of finding a minimum cost perfect matching on S , let M be a minimum cost matching that pairs all the vertices in S except exactly 2. Find an Euler path in $T \cup M$ and shortcut vertices appearing more than once to turn the Euler path into a Hamilton path. Prove or refute: “the described algorithm is a 1.5-approximation algorithm for metric Path-TSP”.

Hint: Recall that if all vertices in a graph have even degree then there exists an Euler tour, and if there are exactly two vertices with odd degree then there exists an Euler path.

— YOUR WORK —